Communication Friction, Managerial Expectation, and Firm Dynamics *

Haoran Wang †

April 7, 2026

Abstract

Large firms are subject to internal communication frictions. This paper develops a theory to incorporate imperfect within-firm communication among input choices into firm dynamics. Firms are modeled as teams of input managers who make decisions based on segmented information. Using this framework, I provide a novel theory-based test to identify communication friction from managerial expectation data: communication frictions allow the capital manager's investment forecast error to simultaneously overreact to past investment (own decision) and underreact to past hiring (other manager's decision), a pattern that cannot be reconciled by alternative mechanisms (e.g. non-convex adjustment costs, behavioral biases). Applying this methodology to managerial forecast data of US public firms confirms this opposite-sign pattern and is consistent with the existence of imperfect communication between capital and labor choices. Quantitatively, communication frictions help match the low capital-labor correlation in the US data. They distort the capital-labor ratio, but they are not a major source of dispersion in capital-labor ratio. Removing communication frictions generates a modest 2.4% gain in aggregate TFP.

*I am most indebted to my advisors, Luminita Stevens and John Haltiwanger, for their continuous guidance and support on this project. I thank Manuel Amador, George-Marios Angeletos, Boragan Aruoba, Bence Bardoczy, Andrew Caplin, Pierre De-Leo, Thomas Drechsel, Emel Filiz-Ozbay, Baris Kaymak, Pawel Krolikowski, Martim Leitao, Chen Lian, Camilo Marchesini, Pablo Ottonello, Benjamin Pugsley, Raffaella Sadun, Karthik Sastry, John Shea and Tao Wang for their helpful comments, and Yifan Shi for the help with the data. I acknowledge the financial support from Summer Research Fellowship, offered by the Graduate School of University of Maryland, College Park.

†Department of Economics, University of Maryland, College Park. Email: hw2688@umd.edu

1 Introduction

As the economy becomes more concentrated over time (Autor, Dorn, Katz, Patterson, and Van Reenen (2020); De Loecker, Eeckhout, and Unger (2020)), frictions faced by large firms become more important in aggregate (Gabaix (2011)). Yet imperfect within-firm communication is one of the large-firm frictions that has been abstracted from macroeconomics studies. Despite strong evidence from micro organizational economics showing that communication frictions abound within large organizations (e.g. Mian (2006); Giroud (2013); Impink, Prat, and Sadun (2024)), macro firm dynamics models typically assume that (potentially noisy) information is perfectly shared among the decision-makers within a firm. As a result, little is known about whether and how communication friction should be modeled in macro, and what aggregate consequences it could generate.

This paper studies the macroeconomic implications of imperfect within-firm communication, leveraging the increasingly available managerial expectation data. Theoretically, I develop a model that incorporates internal communication friction into firms' dynamic choices of capital and labor inputs. To capture communication friction, I model the firm as a team of a capital manager and a labor manager, who are allowed to make their own decisions based on different information sets. A key implication of the model is that the forecast error of capital manager's decision (e.g. investment) can be predicted by both the manager's own past decisions (lagged investment) and the other manager's past decisions (lagged hiring) under communication frictions.

Motivated by this finding, I propose a novel methodology to identify communication friction between capital and labor decisions. The methodology relies on the sign pattern of predictability of investment forecast error by past hiring (other manager's decision) and by past investment (the capital manager's own decision). I show that communication friction is necessary to match the pattern that the two predicatbilities are in opposite direction. In the data of US public firms, I confirm the sign pattern of predictabilities is strongly consistent with the existence of communication friction. In the quantitative exercise, I use this predictability pattern to discipline the parameters of information and communication frictions. When the model is calibrated to match the data, I find that communication frictions contribute significantly to the low investment-hiring correlation in the US data. Our estimates suggests that removing within-firm communication friction generates a modest 2.4% gain in aggregate TFP.

Theory My analysis starts by developing a model that embeds both imperfect information and imperfect communication into an otherwise standard problem of firms choosing capital and labor dynamically. Motivated by Marschak and Radner (1972) and Lian (2020), I model the firm as a team of two managers, one responsible for capital and the other for labor, both of whom dynamically choose their own input to jointly maximize the firm's lifetime profit. To capture communication friction, I allow the managers to observe different private signals about the firm's productivity, so that their information sets are non-nested with each other. The capital and labor choices of the firm, together with the belief dynamics of managers, are characterized by the Markov Perfect Bayesian Equilibrium (MPBE) of this dynamic team production game with incomplete information.

Identifying Communication Friction from Expectation A key prediction of the theory is that communication friction makes the capital manager's investment forecast error predictable by past decisions of both managers. I show that communication friction can make capital manager's investment forecast error overreact to past investment (which is the capital manager's own decision) and underreact to past hiring (which is the other manager's decision) simultaneously, after controlling for sales nowcast errors. This unique opposite-sign pattern of forecast error predictability cannot be easily matched by other behavioral and adjustment frictions, such as extrapolative beliefs (Angeletos, Huo, and Sastry (2021)), cognitive discounting (Gabaix (2020)), diagnostic expectation (Bordalo, Gennaioli, Ma, and Shleifer (2020)), and non-convex adjustment costs. I show analytical and numerically that these alternative mechanisms are much more likely to produce same-sign patterns of forecast error predictability.

The intuition for this result is that under imperfect communication, the predictability by past hiring and past investment convey different messages to the capital manager. Predictability of investment forecast error by past investment reflects how the capital manager utilizes his own past decision, which he knows when he makes the forecast. Absent communication friction, the labor manager's past decision of hiring is based on same information as investment and contains similar information content. Therefore, the predictability by past investment and by past hiring reflect the common frictions or behavioral biases that both managers face and therefore tend to be on the same direction. This symmetry is broken under communication friction: the existence of labor manager's private information makes the capital manager interpret his own past decision and the labor manager's past decision differently, which makes opposite directions of predictability plausible.

I apply this test to the merged IBES/Compustat dataset, which contains both managerial expectation and balance-sheet outcomes for US public firms. I document that after controlling for sales nowcast error, the investment forecast error systematically under-reacts to past hiring (i.e. high past hiring predicts large investment forecast error) but over-reacts to past investment (i.e. high past investment predicts low investment forecast error). This opposite-sign pattern is consistent with the predictions of communication frictions.

Quantification The last part of the paper investigates the implications of communication frictions for firm dynamics and allocative efficiency. I estimate an extended model with adjustment costs, communication frictions and Hsieh-Klenow distortions to jointly match the high serial correlation of input adjustment, low volatility of input adjustment, low investment-hiring correlations, and the opposite-sign pattern in the predictability of investment forecast error in the expectation test. I show that the opposite-sign pattern cannot be matched with other behavioral distortions (such extrapolative beliefs) or non-convex adjustment costs without the help of communication frictions.

Our estimated model suggests that communication friction is a promising source of low investment-hiring correlation in the US data. Under my parameterization, communication friction alone can generate low, even negative investment-hiring correlation in the US data, while none of the other frictions alone can break the perfect correlation between capital and labor adjustments. This is because the existence of private information breaks the perfect correlation between expectations of different managers. From the perspective of Hsieh and Klenow (2009), these manager-level private signals should be regarded as a factor-specific distortion that alters the capital-labor ratio and is a different type of distortion from firm-level, common information, which affects all the inputs symmetrically. Ignoring communication friction will likely overstate the importance of other factor-specific distortions in firm dynamics.

A key difference between communication friction from other behavioral/information frictions under common information is its ability to affect capital-labor ratio. Following the approach by Kaymak and Schott (2023), I provide an analytical decomposition of capital-labor ratio and its dispersion. Communication frictions contribute to the capital-labor ratio by creating a wedge between the managers' beliefs about productivity. However, I show that the overall contribution of communication frictions to the dispersion of capital-labor ratio is quantitatively small. This is because there is a large, negative interaction between communication and adjustment costs that largely offsets the direct contribution of communication frictions to capital-labor ratio dispersion. As a counterfactual, I show that communication frictions can be a more prominent contributor to the dispersion in capital-labor ratio when the adjustment costs are small or strategic complementarity between capital and labor are strong, but its overall contribution is modest.

Communication frictions generate a modest effect on allocative efficiency. Using a similar analytical decomposition, I show that information frictions contribute more to the capital and labor misallocation and can generate larger aggregate TFP losses than adjustment costs and Hsieh-Klenow distortions. I show that the large effect of information frictions results primarily from uncertainty rather than communication frictions: removing communication frictions on top of our baseline calibration slightly lowers the dispersion in MRPK and MRPN and generates a modest TFP gain of 2.4%. This is again because communication frictions can only generate small dispersion in capital-labor ratio. I show that this quantification is robust to non-convex adjustment costs.

Related Literature This paper is related to four strands of literature. First, the theoretical part of the paper is related to the literature that introduces incomplete information into macroeconomic models (Woodford (2001), Angeletos and Lian (2016), Angeletos and Lian (2018), Angeletos and Huo (2021)). My paper makes two contributions to this literature. First, the existing literature has been focusing on introducing information asymmetry across households or firms,[1] while my paper applies this modeling technique to study within-firm frictions. Second, this literature is largely theoretical and does not provide empirical evidence for imperfect common knowledge. My paper provides a novel methodology to empirically identify communication friction from the expectation data and shows evidence in support of communication friction.

Second, the empirical methodology I develop to identify communication friction belongs to the empirical literature that investigates the expectation formation process through the lens of predictability of forecast errors (Coibion and Gorodnichenko (2012, 2015)). Most works in this literature focuses on firm's macroeconomic expectations. Some recent works, notably Ma, Ropele, Sraer, and Thesmar (2020), Barrero (2022) and Asriyan and Kohlhas (2024), study the distortions in the sales expectation of firms, while my paper focuses on investment forecasts. My paper jointly tests the predictability of investment forecast error by past investment and by past hiring and documents simultaneous under- and over-reaction patterns in investment forecast. This is closely related to Kohlhas and Walther (2021), who

[1] An exception is Lian (2021), who studies coordination frictions among multiple decisions of households.

documents simultaneous over- and under-reaction of expectation in a different context.

Third, my work contributes to the organizational economics literature (Malenko (2022)) by investigating how the communication friction affect the managerial expectations. Previous works in this literature document evidence for imperfect communication in two ways. One strand of literature (e.g. Dessein et al. (2022); Impink et al. (2024)) utilizes rich internal datasets that involve highly confidential information such as meeting transcripts, emails and hierarchical details, for a limited number of firms. Such evidence is hard to obtain at a large enough scale for quantifying macroeconomic consequences. In contrast, my methodology relies on managerial expectation, which is increasingly available for a wider range of firms.[2] Another strand of literature proxies communication friction by distance or travel time and reports evidence for a wider range of firms (e.g. Mian (2006); Giroud (2013)). My methodology does not rely on geographical variations and can be more flexibly applied to same-location teams with multiple decision-makers.

Lastly, my paper is closely related to the literature that studies the linkage between information friction/uncertainty and misallocation. Previous work, such as David, Hopenhayn, and Venkateswaran (2016) and David and Venkateswaran (2019), quantifies the contribution of information friction to misallocation under two types of internal information structure: either there is a firm-level noisy signal that is commonly known by the managers, or the labor decision is made ex post and only capital decisions are subject to information frictions. My paper generalizes the analysis and can accommodate any given information structure within the firm. Previous works on uncertainty-driven firm dynamics, such as Bloom, Floetotto, Jaimovich, Saporta-Eksten, and Terry (2018), focus on firm-level uncertainty. My paper shows the importance of subordinates' uncertainty in explaining firm dynamics (which echoes the large within-firm variation in management practices documented by Bloom, Brynjolfsson, Foster, Jarmin, Patnaik, Saporta-Eksten, and Van Reenen (2019)) and highlights the difference between information and communication friction (Bloom, Garicano, Sadun, and Van Reenen, 2014) in firm dynamics.

Layout The remainder of the paper is organized as follows. Section 2 presents the theory. Section 3 describes the empirical test for within-firm communication friction and applies it to U.S. public firms. Section 4 quantifies the aggregate implications of communication frictions, discussing parameter identification and calibration results. Section 5 concludes.

[2]For a recent survey, see Bachmann, Tapa, and van der Klaauw (2023).

2 Model

In this section, I describe the model of dynamic input choices under communication friction. Throughout the paper, there is no aggregate uncertainty, so that all the aggregate variables are fixed at their steady-state values in equilibrium and are common knowledge to all the agents in the economy.

Household There is a representative household in the economy, who has standard preference on final-good consumption, discounts future consumption utility by a discount factor β and supplies a fixed amount of labor $\bar{N}$ to the firms at a wage rate W_t. He can save in a risk-free bond at a real interest rate r in each period. The household chooses how much to save and consume subject to a budget constraint. In a stationary equilibrium, consumption is constant over time, and by the Euler equation, we have $\beta(1+r) = 1$. The only role of the household sector is to close the model, and it is not essential to the analysis.

Final-Good Producer There is a final good producer who combines intermediate goods $\{Y_{it}\}$ into final good Y_t by a CES technology

$$Y_t = \left(\int_0^1 (A_{it} Y_{it})^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}} \tag{1}$$

where A_{it} can be interpreted as an idiosyncratic productivity shock on intermediate-good producer i or a demand shock on the product i, and $\epsilon > 1$ is the parameter for elasticity of substitution.

The price of final good is normalized to one. Given price $\{P_{it}\}$, the final good producers choose how much intermediate goods to purchase from intermediate-good producers. This yields a standard demand curve

$$P_{it} = \left(\frac{Y_t}{Y_{it}} \right)^{\frac{1}{\epsilon}} A_{it}^{1-\frac{1}{\epsilon}} \tag{2}$$

Intermediate-Good Producer There is a continuum $i \in [0, 1]$ of intermediate-good producers in the economy. Each producer i has a constant-return-to-scale production function to combine capital and labor inputs into intermediate goods:

$$Y_{it} = K_{it}^{\alpha} N_{it}^{1-\alpha} \tag{3}$$

where $\alpha \in (0,1)$ is the parameter for capital share. Each producer i operates in a standard monopolistic competition environment: it internalizes the demand function (2), so that its revenue can be expressed as a decreasing-return-to-scale function of capital and labor, denoted as $R(K_{it}, N_{it}; A_{it})$:

$$R(K_{it}, N_{it}; A_{it}) = P_{it} Y_{it} = Y_t^{\frac{1}{\epsilon}} A_{it}^{1-\frac{1}{\epsilon}} Y_{it}^{1-\frac{1}{\epsilon}} = Y_t^{\frac{1}{\epsilon}} A_{it}^{1-\frac{1}{\epsilon}} K_{it}^{\hat{\alpha}_1} N_{it}^{\hat{\alpha}_2} \tag{4}$$

where $\hat{\alpha}_1 \equiv \alpha(1 - 1/\epsilon)$ and $\hat{\alpha}_2 \equiv (1-\alpha)(1-1/\epsilon)$.

Producer i's cost of production includes two parts. First, the firm pays the rental cost of capital and hiring cost of labor. Additionally, the firm faces convex adjustment costs

$$\frac{\xi_k}{2}\left(\frac{I_{it}}{K_{i,t-1}} - \delta\right)^2 K_{i,t-1} \text{ and } \frac{\xi_n}{2}\left(\frac{H_{it}}{N_{i,t-1}}\right)^2 N_{i,t-1}$$

where $I_{it} \equiv K_{it} - (1-\delta)K_{i,t-1}$ is investment and $H_{it} \equiv N_{it} - N_{i,t-1}$ is hiring. ξ_k and ξ_n are parameters that control the slope of marginal adjustment cost for capital and labor respectively. Putting everything together, the cost function $\mathcal{C}(K_{it}, N_{it}; K_{i,t-1}, N_{i,t-1})$ is

$$\mathcal{C}(K_{it}, N_{it}; K_{i,t-1}, N_{i,t-1}) = R_t K_{it} + W_t N_{it} + \frac{\xi_k}{2}\left(\frac{I_{it}}{K_{i,t-1}} - \delta\right)^2 K_{i,t-1} + \frac{\xi_n}{2}\left(\frac{H_{it}}{N_{i,t-1}}\right)^2 N_{i,t-1}$$

where δ is the deprecation rate, and R_t is the rental rate of capital, which is equal to $r + \delta$ in a stationary equilibrium.

Denote profit function $\Pi(K_{it}, N_{it}; K_{i,t-1}, N_{i,t-1}, A_{it}) \equiv R(K_{it}, N_{it}; A_{it}) - \mathcal{C}(K_{it}, N_{it}; K_{i,t-1}, N_{i,t-1})$. Under full information, the firm's input choices are characterized by the Bellman equation

$$V(K_{i,t-1}, N_{i,t-1}; A_{it}) = \max_{K_{it}, N_{it}} \Pi(K_{it}, N_{it}; K_{i,t-1}, N_{i,t-1}, A_{it}) + \frac{1}{1+r}\mathbb{E}[V(K_{it}, N_{it}; A_{i,t+1})|A_{it}] \tag{5}$$

To facilitate my later discussions, it is useful to think of the firm as a team of two managers, one in charge of capital decisions (I call him the Capital Manager) and the other in charge of labor decisions (I call her the Labor Manager), who share the common objective of maximizing the firm's lifetime profit. Hence, the capital and labor policy function that solves the *firm's* Bellman equation (5) are equivalent to a Markov Perfect Equilibrium (MPE) of a dynamic team production game (Marschak and Radner (1972), Chapter 7), in which

- Given the labor manager's policy N_{it}, the capital manager's strategy K_{it} is the one that solves the Bellman equation

$$V^k(K_{i,t-1}, N_{i,t-1}; A_{it}) = \max_K \Pi(K, N_{it}; K_{i,t-1}, N_{i,t-1}, A_{it}) + \frac{1}{1+r}\mathbb{E}[V^k(K, N_{it}; A_{i,t+1})|A_{it}] \tag{6}$$

- Given the capital manager's policy K_{it}, the capital manager's strategy N_{it} is the one that solves the Bellman equation

$$V^n(K_{i,t-1}, N_{i,t-1}; A_{it}) = \max_N \Pi(K_{it}, N; K_{i,t-1}, N_{i,t-1}, A_{it}) + \frac{1}{1+r}\mathbb{E}[V^n(K_{it}, N; A_{i,t+1})|A_{it}] \tag{7}$$

Equation (6) and (7) are the *manager*-level Bellman equations that characterize the firm's input decisions. This reformulation gives us more flexibility to accommodate incomplete/asymmetric information between managers within the firm and makes it easier for us to analyze a richer internal information structure within the firm.

Information The only source of uncertainty in this simple model is the idiosyncratic firm fundamentals A_{it}. I assume that the log of A_{it}, denoted as a_{it}, follows an AR(1) process

$$a_{it} = \rho a_{i,t-1} + \mu_{it}, \text{ with } \mu_{it} \overset{iid}{\sim} N(0, \sigma_\mu^2) \tag{8}$$

The firm managers know the data generating process in (8) and treat the objective distribution of μ_{it} or a_{it} as the prior. However, the true value of a_{it} is never fully revealed to the firm and its managers.[3] Hence, when making decisions, the capital and labor manager of firm i need to learn dynamically from the Gaussian noisy signals they receive to form expectations about the "latent variable" a_{it} over time using Kalman filters, which I will specify later in detail.

The key friction is that I allow each manager's decision to be dependent on different, non-nested information sets. Formally, let $\mathcal{I}_{it}^k$ denote capital manager's information set in period

[3] This is different from the setup in David, Hopenhayn, and Venkateswaran (2016) and David and Venkateswaran (2019), where the lagged firm fundamental $a_{i,t-1}$ is fully revealed to the firm in period t.

t, and $\mathcal{I}_{it}^n$ denote the labor manager's information set in period t. In particular, I allow the possibility that $\mathcal{I}_{it}^k \neq \mathcal{I}_{it}^n$ and are not a subset to each other. This captures the notion of imperfect communication within the organization: with imperfect information sharing, the managers' information sets do not perfectly overlap.

I now specify the elements in the managers' information sets $\mathcal{I}_{it}^k$ and $\mathcal{I}_{it}^n$. I assume that there is a piece of firm-level information s_{it}^p, which is a noisy signal of the firm's fundamental a_{it} and is commonly known by the managers (i.e. $s_{it}^p \in \mathcal{I}_{it}^k \cup \mathcal{I}_{it}^n$):

$$s_{it}^p = a_{it} + \epsilon_{it}^p, \text{ with } \epsilon_{it}^p \overset{iid}{\sim} N(0, \sigma_{\epsilon,p}^2) \tag{9}$$

For the non-nested part of the information set, I assume that each manager $m \in \{k, n\}$ is endowed with a private, noisy signal a_{it}^m about a_{it}:

$$a_{it}^m = a_{it} + \epsilon_{it}^m, \text{ with } \epsilon_{it}^m \overset{iid}{\sim} N(0, \sigma_{\epsilon,m}^2) \text{ for } m \in \{k, n\}. \tag{10}$$

The signal noises $\epsilon_{it}^k, \epsilon_{it}^n, \epsilon_{it}^p$ and the innovation in the firm fundamental μ_{it} are independent to each other. In each period, the realization of s_{it}^k and s_{it}^n are private to the capital and labor manager respectively, but the data generating process of the signals, (9) and (10), are commonly known by the managers.

The standard deviations of the signal noises, $(\sigma_{\epsilon,p}, \sigma_{\epsilon,k}, \sigma_{\epsilon,n})$, define the internal information structure of the firm and are the key parameters to pin down in the quantitative analysis. This parametrization gives us a parsimonious way to model the degree of information segmentation within the firm. For example, if we make the private signals to be extremely noisy, i.e. $\sigma_{\epsilon,m} \to \infty$ for $m \in \{k, n\}$, then the managers will put essentially zero weight on private signals when they apply Bayes's rule to update their beliefs, and the firm's behavior is as if the firm is endowed with a public information only and the managers' information set perfectly overlaps. On the other hand, if $\sigma_{\epsilon,p} \to \infty$ while $\sigma_{\epsilon,m} > 0$, then the managers will not rely on the firm-level signal to update their beliefs, and it's as if the firm is operating on a non-nested information structure.

Since the realization of signal a_{it}^m is private to manager m and both managers choose their decisions simultaneously, the action that manager m takes in period t is not known by the other manager m', and vice versa. However, I assume that at the end of each period t, the actions taken by the managers in period t, K_{it} and N_{it}, are revealed to both managers and become public information. This assumption allows the managers to use the observed actions in the previous period to update their beliefs about the unknown fundamental a_{it}

and the other manager's action in the current period, which simplifies the characterization of the equilibrium concept. Formally, putting everything together, we can express the law of motion for the managers' information sets $\mathcal{I}^k_{it}$ and $\mathcal{I}^n_{it}$ as

$$\mathcal{I}^k_{it} = \mathcal{I}^k_{i,t-1} \cup \{s^p_{it}, s^k_{it}, N_{i,t-1}\} \text{ and } \mathcal{I}^n_{it} = \mathcal{I}^n_{i,t-1} \cup \{s^p_{it}, s^n_{it}, K_{i,t-1}\} \tag{11}$$

Equilibrium Under communication friction, the firm faces a dynamic team production problem with incomplete information. The firm's input decisions can therefore be interpreted as a Markov Perfect Bayesian Equilibrium (MPBE), in which

- Given labor manager's policy function $N_{it}(\cdot)$, capital manager's policy function $K_{it} : \mathcal{I}^k_{it} \to \mathbb{R}^+$ solves Bellman equation

  $$V^k(K_{i,t-1}, N_{i,t-1}; \mathcal{I}^k_{it}) = \max_K \mathbb{E}\left[\Pi(K, N_{it}(s^n_{it}); K_{i,t-1}, N_{i,t-1}, A_{it}) + \beta V^k(K, N_{it}(s^n_{it}); \mathcal{I}^k_{i,t+1}) \middle| \mathcal{I}^k_{it}\right] \tag{12}$$

  where s^n_{it} is the signal realization of the labor manager in period t, and

- Given capital manager's policy function $K_{it}(\cdot)$, labor manager's policy function $N_{it} : \mathcal{I}^n_{it} \to \mathbb{R}^+$ solves Bellman equation

  $$V^n(K_{i,t-1}, N_{i,t-1}; \mathcal{I}^n_{it}) = \max_N E_t\left[\Pi(K_{it}(s^k_{it}), N; K_{i,t-1}, N_{i,t-1}, A_{it}) + \beta V^n(K_{it}(s^k_{it}), N; \mathcal{I}^n_{i,t+1}) \middle| \mathcal{I}^n_{it}\right] \tag{13}$$

  where s^k_{it} is the signal realization of the capital manager in period t, and

- In period t, each manager updates his/her belief about a_{it} using his/her observed signals and the past actions $K_{i,t-1}, N_{i,t-1}$, consistent with the policy function $K_{it}(\cdot), N_{it}(\cdot)$ and the Bayes's rule.

The advantage of using MPBE in (12) and (13) to characterize firm's behavior lies in its flexibility of accommodating essentially any possible internal information structure of the firm. This flexibility makes it a more general model and nests two classes of extensively studied models in the firm dynamics literature. One is the models based on full-information rational expectation (FIRE), as in (5) or (6) and (7) above. They can be regarded as a special case of (12) and (13), with $\mathcal{I}^k_{it} = \mathcal{I}^n_{it} = \{a_{it}, a_{i,t-1}, \cdots, a_{i0}\}$, i.e. the true firm fundamental is always revealed to both managers. Using our parameterization (9) and (10) above, this full-information benchmark maps to the limit $\sigma_{\epsilon,p} \to 0$ and $\sigma_{\epsilon,m} \to \infty$ for $m \in \{k, n\}$. Another class of models are the ones based on common, noisy information, as in David, Hopenhayn, and Venkateswaran (2016).[4] Although it is a useful benchmark to analyze the effect of uncertainty and imperfect information on firm dynamics, it presumes that that imperfectly observed information is frictionlessly shared and communicated within the organization. From the above discussion, this is again a special case of the current framework, with $\mathcal{I}_{it}^k = \mathcal{I}_{it}^n = \{s_{it}^p, \cdots, a_{i0}^p\}$. Using our parameterization (9) and (10) above, this noisy, common information benchmark maps to the limit $\sigma_{\epsilon,p} \in (0, \infty)$ and $\sigma_{\epsilon,m} \to \infty$ for $m \in \{k, n\}$. Hence, the firm dynamics implied by noisy, common information models is also contained in (12) and (13).

Equation (12) and (13) illustrate that the managers face two types of uncertainty when they are making their own decisions. For the capital manager, apart from the fundamental uncertainty about a_{it} or A_{it}, he faces a strategic uncertainty about the action of the labor manager N_{it}, which is measurable to labor manager's information set only. The source of this strategic uncertainty is the fact that information is not perfectly shared within the firm, so that there is labor-specific information that is only known to the labor manager and not to the capital manager. This strategic uncertainty affects capital manager's decision because labor manager's action affects both the marginal revenue product of capital and the continuation value of the capital manager, and therefore capital manager has an incentive to form expectation about N_{it} to guide his own decision about K_{it}.

Log-Quadratic Approximation The third requirement of an MPBE requires that the belief dynamics of both managers are consistent with the policy function and Bayes's rule. It is a difficult task to analytically characterize or compute the evolution of beliefs since it is an infinite-dimensional object.

Here, I get around this technical challenge by doing a Woodford (2003)-type log-quadratic approximation on the profit function with respect to $(K_{it}, N_{it}, K_{i,t-1}, N_{i,t-1}, A_{it})$ around their frictionless steady-state values.[5] I can therefore express the log-quadratic approximated profit function π_{it} as a quadratic function in terms of the log deviations of capital, labor and firm fundamental from their own frictionless steady-state value. Denote these log-deviation

[4] More precisely, David et al. (2016) studies two extreme cases, where (1) imperfectly observed information is perfectly shared among capital and labor managers, and (2) labor decisions are made *ex post* and frictionlessly, and only capital decisions are subject to information friction (which is also the main specification of David and Venkateswaran (2019)). The latter case is also a special case of the current framework, with the capital manager's information set is a subset of the labor manager's information set. My paper can be regarded as providing a smooth version of information friction between these two extreme cases.

[5] By frictionless, I mean the case where there is no information friction and adjustment friction, i.e. a_{it} is fully revealed to the firm, and $\xi_k = \xi_n = 0$.

values as k_{it}, n_{it} and a_{it} respectively, and define investment (rate) as $\iota_{it} \equiv k_{it} - k_{i,t-1}$ and hiring (rate) as $h_{it} \equiv n_{it} - n_{i,t-1}$. The approximated profit function can be written as

$$\pi(\iota_{it}, h_{it}; x_{it}) = x_{it}'Px_{it} + x_{it}'Q\iota_{it} + x_{it}'Rh_{it} + H_\iota \iota_{it}^2 + H_{\iota h}\iota_{it}h_{it} + H_h h_{it}^2 \tag{14}$$

where $x_{it} = [k_{i,t-1}, n_{i,t-1}, a_{it}]'$ is the state vector of the firm with a law of motion

$$x_{i,t+1} = Ax_{it} + B\iota_{it} + Ch_{it} + D\mu_{i,t+1} \tag{15}$$

and matrices $A, B, C, D, P, Q, R, H_\iota, H_h$ and $H_{\iota h}$ are constants imputed from the frictionless steady-state values and model parameters. The derivations of (14) are left to Appendix A.1.

Characterization of Linear MPBE The log-quadratic approximation of profit function allows us to characterize a linear MPBE analytically.

Plugging (14) and (15) back into the manager-level Bellman equations (12) and (13) transforms the original problem into a linear-quadratic (LQ) control problem, with investment ι_{it} and hiring h_{it} as control variables and x_{it} as the state vector. An immediate implication of the LQ control is that the policy function is linear and certainty equivalence holds:

Lemma 1. *With the log-quadratic-approximated profit function (14) and the law of motion for state (15), we have*

1. *The policy function for the full-information MPE in (6) and (7) takes the form*

$$\begin{aligned} \iota_{it} &= F_\iota x_{it} = F_\iota^k k_{i,t-1} + F_\iota^n n_{i,t-1} + F_\iota^a a_{it} \\ h_{it} &= F_h x_{it} = F_h^k k_{i,t-1} + F_h^n n_{i,t-1} + F_h^a a_{it} \end{aligned}$$

   where the policy matrix F_ι, F_h can be solved by iterating forward a pair of Ricatti equations;

2. *The policy function for (12) and (13) under communication friction takes the form*

$$\iota_{it} = F_\iota \mathbb{E}[x_{it}|\mathcal{I}_{it}^k] = F_\iota^k k_{i,t-1} + F_\iota^n n_{i,t-1} + F_\iota^a \hat{a}_{it}^k \tag{16}$$

$$h_{it} = F_h \mathbb{E}[x_{it}|\mathcal{I}_{it}^n] = F_h^k k_{i,t-1} + F_h^n n_{i,t-1} + F_h^a \hat{a}_{it}^n \tag{17}$$

   where F_ι, F_h are the same policy matrix as in the full-information MPE, and $\hat{a}_{it}^k \equiv \mathbb{E}[a_{it}|\mathcal{I}_{it}^k]$ and $\hat{a}_{it}^n \equiv \mathbb{E}[a_{it}|\mathcal{I}_{it}^n]$ are the posterior mean of capital and labor manager's *first-order belief on a_{it}.*

Proof. See Appendix A.3. □

The linearity of (16) and (17), together with the Gaussian structure of μ_{it} and signal noises, significantly simplifies the characterization of the belief updating process for two reasons. First, the Gaussian shocks and noises imply that tracking the first and second moment is sufficient for the characterization of belief dynamics, which significantly lowers the dimensions needed for the computation. Moreover, the linearity of policy functions makes actions jointly Gaussian with firm fundamentals and signals, and the managers can simply treat the other manager's lagged action as a noisy signal of the firm fundamentals and learn from it in the same manner as from other signals. For instance, in period t, with (17), the capital manager forms belief about the labor manager's action h_{it} by

$$\hat{h}^k_{it} \equiv \mathbb{E}[h_{it}|\mathcal{I}^k_{it}] = F^k_h k_{i,t-1} + F^n_h n_{i,t-1} + F^a_h \mathbb{E}[\hat{a}^n_{it}|\mathcal{I}^k_{it}] \tag{18}$$

Equation (18) conveys two important messages. First, it tells us that the capital manager's belief about the labor manager's action is linear on his *high-order belief* about labor manager's nowcast on a_{it}, and therefore it is necessary for him to track the evolution of both first-order beliefs $a_{it}\big|\mathcal{I}^k_{it} \sim N(\hat{a}^k_{it}, \hat{\Sigma}^k_t)$ and high-order beliefs $\hat{a}^n_{it}\big|\mathcal{I}^k_{it} \sim N(\hat{a}^{(k,n)}_{it}, \hat{\Sigma}^{(k,n)}_t)$ along the belief updating process.[6] Second, it tells us that the elasticity of hiring to firm fundamental, F^a_h, directly enters the belief formation process by affecting the nowcast error $h_{it} - \hat{h}^k_{it}$. This channel of policy matrix affecting beliefs will be in effect if and only if there is information asymmetry between the managers: if the information is symmetric within the firm, the capital manager knows that the labor manager shares the same information with him, which allows him to know the labor manager's action exactly, i.e. $h_{it} = \hat{h}^k_{it}$. Similarly, the labor manager updates her beliefs using capital manager's past action and policy function (16) in a MPBE. For that purpose, she will need to track her first-order belief $a_{it}\big|\mathcal{I}^n_{it} \sim N(\hat{a}^n_{it}, \hat{\Sigma}^n_t)$ and high-order belief $\hat{a}^k_{it}\big|\mathcal{I}^n_{it} \sim N(\hat{a}^{(n,k)}_{it}, \hat{\Sigma}^{(n,k)}_t)$ dynamically.

The following lemma characterizes the evolution of the posterior mean $\hat{a}^k_{it}, \hat{a}^{(k,n)}_{it}$ and $\hat{a}^n_{it}, \hat{a}^{(n.k)}_{it}$ for a given pair of policy matrices (F_ι, F_h).

Lemma 2. *For a given pair of policy matrices F_ι and F_h, the posterior first-order and*

[6] Here, the second-order mean $\hat{a}^{k,n}$ is the capital manager's perception of labor manager's beliefs, and $\hat{a}^{(k,n)} \equiv \mathbb{E}[\hat{a}^n|\mathcal{I}^k]$.

high-order mean $\hat{a}^k_{it}, \hat{a}^{(k,n)}_{it}$, $\hat{a}^n_{it}, \hat{a}^{(n.k)}_{it}$ evolves by

$$\begin{aligned} \hat{a}^k_{it} &= \hat{a}^{(k,n)}_{it} = (\hat{a}^k_{it})^- + G^k_t \left(z^k_{it} - H(\hat{a}^k_{it})^-\right) && (19) \\ \hat{a}^n_{it} &= \hat{a}^{(n,k)}_{it} = (\hat{a}^n_{it})^- + G^n_t \left(z^n_{it} - H(\hat{a}^n_{it})^-\right) && (20) \end{aligned}$$

where $z^k_{it} = [s^p_{it}, s^k_{it}]'$, $z^n_{it} = [s^p_{it}, s^n_{it}]'$, $H = [1, 1]'$, with the Kalman gains being

$$G^k_t = (\hat{\Sigma}^k_t)^- H' \left[H(\hat{\Sigma}^k_t)^- H' + W^k\right]^{-1} \text{ and } G^n_t = (\hat{\Sigma}^n_t)^- H' \left[H(\hat{\Sigma}^n_t)^- H' + W^n\right]^{-1}$$

and the pre-estimates $(\hat{a}^k_{it})^-, (\hat{a}^n_{it})^-$ are given by

$$\begin{aligned} (\hat{a}^k_{it})^- &= \rho(\hat{a}^k_{i,t-1})^+ = \rho\left[\hat{a}^k_{i,t-1} + J^k_{t-1}(h_{i,t-1} - \hat{h}^k_{i,t-1})\right] && (21) \\ (\hat{a}^n_{it})^- &= \rho(\hat{a}^n_{i,t-1})^+ = \rho\left[\hat{a}^n_{i,t-1} + J^n_{t-1}(\iota_{i,t-1} - \hat{\iota}^n_{i,t-1})\right] && (22) \end{aligned}$$

with the Kalman gain being

$$J^k_{t-1} = (\hat{\Sigma}^k_{t-1} - \tilde{\Sigma}^{(k,n)}_{t-1}) \left[\hat{\Sigma}^{(k,n)}_{t-1} F^a_h\right]^{-1} \text{ and } J^n_{t-1} = (\hat{\Sigma}^n_{t-1} - \tilde{\Sigma}^{(k,n)}_{t-1}) \left[\hat{\Sigma}^{(k,n)}_{t-1} F^a_\iota\right]^{-1}$$

The evolution of the first-order and higher order uncertainty $(\hat{\Sigma}^k_t, \hat{\Sigma}^n_t, \hat{\Sigma}^{(k,n)}_t)$ is defined by a triplet of Riccati equations (A5), (A6) and (A7) in Appendix A.2.

Proof. See Appendix A.2. □

In Lemma 2, the belief updating from period $t-1$ to t is done in two steps. In the first step, the managers observe the other manager's action in the previous period, $h_{i,t-1}$ and $\iota_{i,t-1}$, and use them to form a pre-estimate $(\hat{a}^k_{it})^-$ and $(\hat{a}^n_{it})^-$, according to (21) and (22). In the second step, the managers use the signal realization in period t to update their beliefs on top of these pre-estimates, following equation (19) and (20). Note that the first step of belief updating exists if and only if the managers' information sets are non-nested. Absent asymmetric information, there would be no strategic uncertainty between managers, so that $h_{i,t-1} = \hat{h}^k_{i,t-1}$ and the equation (21) degenerates to $(\hat{a}^k_{it})^- = \rho\hat{a}^k_{i,t-1}$. In that case, the labor manager's action $h_{i,t-1}$ is already contained in the capital manager's information set in period $t-1$, and hence revealing $h_{i,t-1}$ to the capital manager doesn't give him additional information about the unobserved firm fundamental.

Now we have all the ingredients to characterize a linear MPBE associated with (14) and (15).

Lemma 3. *A linear Markov Perfect Bayesian Equilibrium is a set of policy* (F_ι, F_h) *and beliefs* $(\hat{a}^k_{it}, \hat{\Sigma}^k_t)$, $(\hat{a}^n_{it},, \hat{\Sigma}^n_t)$, $(\hat{a}^{(k,n)}_{it}, \hat{\Sigma}^{(k,n)}_t)$, *and* $(\hat{a}^{(n,k)}_{it}, \hat{\Sigma}^{(n,k)}_t)$ *such that*

- *Beliefs are consistent with strategies: Given policy* (F_ι, F_h)*, the capital and labor managers' update their beliefs* $(\hat{a}^k_{it}, \hat{\Sigma}^k_t)$, $(\hat{a}^n_{it}, \hat{\Sigma}^n_t)$, $(\hat{a}^{(k,n)}_{it}, \hat{\Sigma}^{(k,n)}_t)$, *and* $(\hat{a}^{(n,k)}_{it}, \hat{\Sigma}^{(n,k)}_t)$ *by the Bayes rule, according to Equations (19) to (20) in Lemma 2;*
- *Strategies are consistent with beliefs: Given the beliefs* $(\hat{a}^k_{it}, \hat{\Sigma}^k_t)$, $(\hat{a}^n_{it},, \hat{\Sigma}^n_t)$, $(\hat{a}^{(k,n)}_{it}, \hat{\Sigma}^{(k,n)}_t)$, *and* $(\hat{a}^{(n,k)}_{it}, \hat{\Sigma}^{(n,k)}_t)$*, the policy matrices* (F_ι, F_h) *solves the Bellman equations (6) and (7), and the policy function for investment and hiring satisfy (16) and (17), as in Lemma 1.*

The linear MPBE in Lemma 3 implies a law of motion for the joint distribution of action, expectation and state, namely, the vector $\equiv [k_{it}, n_{it}, \hat{a}^k_{it}, \hat{a}^n_{it}, a_{it}]'$. A stationary equilibrium requires this distribution to be at its stationary distribution. The computation of this stationary distribution is essential to the quantitative analysis in Section 4 and the details of computation are left to Appendix A.4.

3 Identifying Communication Friction

In this section, I describe the empirical strategy of identifying communication friction from managerial expectation data and implement the methodology to U.S. public firms.

3.1 Communication Frictions and Managerial Expectations

The key object that maps the theory to data is the investment forecast error, defined as the difference between actual capital investment ι_{it} and the capital manager's one-period-ahead forecast of capital investment, $\mathbb{F}^k_{i,t-1}[\iota_{it}]$. Note that the expectation operator $\mathbb{F}^k_{i,t-1}$ is allowed to deviate from rational expectation, but for now let's assume that $\mathbb{F}^k_{i,t-1}$ is the same as rational expectation $\mathbb{E}^k_{i,t-1} \equiv \mathbb{E}[\cdot|\mathcal{I}^k_{i,t-1}]$, in the sense that the capital manager updates his beliefs following the Bayes's rule with correct perception of model parameters (such as the persistence of productivity ρ). From the linear investment decision rule (16) and the belief updating equation (19) and (21) for the capital manager, we have the following proposition that characterize the structure of investment forecast error:

Proposition 1. *In a MPBE defined in Lemma 3, the investment forecast error of firm i takes the form*

$$\iota_{it} - \mathbb{E}^k_{i,t-1}[\iota_{it}] = \beta_1(a_{i,t-1} - \hat{a}^k_{i,t-1}) + \beta_2(h_{i,t-1} - \hat{h}^k_{i,t-1}) + \beta_3(H\mu_{it} + w^k_{it}) \tag{23}$$

where $\beta_1, \beta_2 \in \mathbb{R}$ and $\beta_3 \in \mathbb{R}^{1\times 2}$ are constants in terms of the policy matrices F_ι, F_h as in Lemma 1 and Kalman gain matrices G^k, G^n, J^k, J^n in Lemma 2.

Proof. See Appendix A.5. □

Proposition 1 shows that the investment forecast error has three components. The first term captures the fundamental uncertainty that the capital manager faces. It is the gap between actual productivity $a_{i,t-1}$ and the capital manager's expectation in period $t-1$ about the productivity in that period. As long as the capital manager is uncertain about the true productivity in period $t-1$, this gap will not be zero. The third term $\beta_3(H\mu_{it} + w^k_{it})$ collects all the time-t productivity innovations and signal noise, which are orthogonal to information on or before period $t-1$.

The second term, $\beta_2(h_{i,t-1} - \hat{h}^k_{i,t-1})$, speaks directly to the communication friction. It is the gap between actual hiring in period $t-1$ and the capital manager's expectation in period $t-1$ about $h_{i,t-1}$. Absent communication friction, this term will always equal to 0, as the capital manager knows that the labor manager's decision is based on the same information as his, so that he has no problem of knowing the labor manager's decision in $t-1$. His expectation $\hat{h}^k_{i,t-1}$ will therefore be exactly the same as $h_{i,t-1}$. In that case, the model degenerates to the standard noisy-information environment, as in Angeletos and Lian (2016): the investment forecast error can only be predicted by the productivity nowcast error, which reflects the manager's uncertainty in productivity. If communication is frictional within the organization, the capital manager does not observe the exact $h_{i,t-1}$ in period $t-1$ because of the labor manager's private information, and as a result, his belief about the labor manager's action in that period is generally not the same as what the labor manager's actual decision, making this strategic uncertainty contribute to the investment forecast error.

Equation (23) is the key equation that motivates the empirical strategy of identifying communication frictions. Before taking the model to the data, I make one additional manipulation to (23). Since the productivity expectations are not available in the data, I use the capital manager's sales nowcast error $y_{i,t-1} - \hat{y}^k_{i,t-1}$ to replace $a_{i,t-1} - \hat{a}^k_{i,t-1}$ in (23). Following from the Cobb-Douglas revenue function (4), the logged sales $y_{i,t-1}$ satisfy

$$y_{i,t-1} = (1 - 1/\epsilon)a_{i,t-1} + \hat{\alpha}_1 k_{i,t-1} + \hat{\alpha}_2 n_{i,t-1}$$

and therefore the capital manager's sales nowcast error in period $t-1$ can be written as

$$y_{i,t-1} - \hat{y}^k_{i,t-1} = (1 - 1/\epsilon)(a_{i,t-1} - \hat{a}^k_{i,t-1}) + \hat{\alpha}_2(h_{i,t-1} - \hat{h}^k_{i,t-1})$$

i.e. it is a linear combination of the productivity nowcast error and the hiring nowcast error. Plug it into (23), we get a model-implied relationship between investment forecast error, lag sales nowcast error, lag hiring nowcast error, and time-t innovations/noise:

$$\iota_{it} - \mathbb{E}^k_{i,t-1}[\iota_{it}] = \beta_h(h_{i,t-1} - \hat{h}^k_{i,t-1}) + \beta_y(y_{i,t-1} - \hat{y}^k_{i,t-1}) + \nu_{it} \tag{24}$$

Equation (24) makes it possible for us to utilize the managerial expectations on capital investment and sales to test the communication frictions within the firm.

3.2 Identification Strategy

Equation (23) and (24) imply that, in presence of productivity uncertainty and communication frictions, investment forecast error can be predicted by past variables. I consider the following two regressions:

$$\iota_{it} - \mathbb{E}^k_{i,t-1}[\iota_{it}] = \tilde{\beta}_h h_{i,t-1} + \beta_y(y_{i,t-1} - \hat{y}^k_{i,t-1}) + \nu_{it} \tag{25}$$

and

$$\iota_{it} - \mathbb{E}^k_{i,t-1}[\iota_{it}] = \tilde{\beta}_\iota \iota_{i,t-1} + \beta_y(y_{i,t-1} - \hat{y}^k_{i,t-1}) + \nu_{it} \tag{26}$$

The coefficients of interest are $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$. The following result characterizes the relationship between these regression coefficients and communication frictions within the firm.

Proposition 2. *The regression coefficients $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ have the following properties:*

1. *Without communication frictions, $\tilde{\beta}_h = \tilde{\beta}_\iota = 0$.*

2. *Under communication frictions, $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ need not be zero. In particular, communication frictions can generate an opposite-sign pattern: $\tilde{\beta}_h > 0$ and $\tilde{\beta}_\iota < 0$.*

To prove and illustrate intuition for this result, following Frisch-Waugh-Lowell theorem and equation (24), I express the regression coefficient $\tilde{\beta}_h$ as

$$\tilde{\beta}_h = \beta_h \frac{cov[M_y(h_{i,t-1} - \hat{h}^k_{i,t-1}), M_y h_{i,t-1}]}{var(M_y h_{i,t-1})} \tag{27}$$

where M_y is the residual matrix of sales nowcast error $y_{i,t-1} - \hat{y}^k_{i,t-1}$. The residualized hiring surprise $\beta_h M_y(h_{i,t-1} - \hat{h}^k_{i,t-1})$ is the component of investment forecast error that cannot be explained by the sales nowcast error and correlates with hiring in $t-1$. The coefficient regression $\tilde{\beta}_h$ therefore captures whether the revelation of lagged hiring provides additional information to the capital manager beyond the firm-level news reflected in sales surprise. Apparently, without communication friction, lagged hiring is contained in the capital manager's information set in period $t-1$, and the hiring surprise $h_{i,t-1} - \hat{h}^k_{i,t-1}$ is always 0, making $\tilde{\beta}_h = 0$. Having a nonzero $\tilde{\beta}_h$ distinguishes communication frictions from models with common, noisy information, no matter whether the signal noise is exogenous (as in David et al. (2016)) or endogenously generated from costly information acquisition (as in Gabaix (2014)). It also distinguishes communication frictions from mechanisms that can be nested in the current setup (with approximated profit function (14) and (15)), such as convex adjustment costs, time-to-build and Hsieh-Klenow-type distortions: none of them can generate a nonzero $\tilde{\beta}_h$ by themselves without communication frictions.

Allowing for communication frictions makes it possible for $\tilde{\beta}_h$ to be nonzero. Assuming parameter $\beta_h > 0$, a positive regression coefficient $\tilde{\beta}_h$ means the capital manager's expectation in period $t-1$ is *underreacting* to hiring. The intuition is in the same spirit of Coibion and Gorodnichenko (2015): because of communication frictions, labor manager's hiring decision in period $t-1$ is not fully internalized by the capital manager in period $t-1$, and therefore the investment forecast of the capital manager $\mathbb{E}^k_{i,t-1}[\iota_{it}]$ does not fully incorporate hiring. The labor information will only be available to the capital manager in period t and will be reflected in the actual investment ι_{it}. This creates the positive predictability of investment forecast error by past hiring.

Similarly, the regression coefficient $\tilde{\beta}_\iota$ can be expressed as

$$\tilde{\beta}_\iota = \beta_h \frac{cov[M_y(h_{i,t-1} - \hat{h}^k_{i,t-1}), M_y \iota_{i,t-1}]}{var(M_y \iota_{i,t-1})} \tag{28}$$

Without communication friction, the residualized hiring surprise is always 0, making $\tilde{\beta}_\iota = 0$. Communication frictions can lead $\tilde{\beta}_\iota$ to be nonzero. Furthermore, it is possible for the regression coefficient $\tilde{\beta}_\iota$ to be negative under communication frictions if the correlation between residualized hiring surprise and investment is negative. This negative correlation is plausible when the firm-level public information is noisy: in that case, the capital manager's decision and expectation in period $t-1$ are mainly driven by his private information, which is not fully internalized by the labor manager in her labor decision $h_{i,t-1}$. If a high investment $\iota_{i,t-1}$ is due to a high realization of capital manager's private signal, it is likely to associate with an optimistic hiring expectation $\hat{h}^k_{i,t-1}$ by the capital manager and a low actual hiring from the labor manager, generating a small or even negative hiring surprise. This mechanism makes it possible for the correlation between hiring nowcast error and investment to be negative.

3.3 Sign Patterns under Alternative Mechanisms

The potential opposite-sign pattern established in Proposition 2 is a unique property of communication frictions that cannot be easily reproduced by other behavioral and adjustment frictions that can generate a nonzero coefficient in $\tilde{\beta}_h$ and/or $\tilde{\beta}_\iota$. To illustrate this, I mute communication frictions between managers for now and analyze the sign pattern of regression coefficients under a few alternative mechanisms.

Time-to-build, Reduced-form Distortions, and Private Benefits I first consider the sign patterns of regression (25) and (26) under three alternative mechanisms: time-to-build in capital and/or labor, Hsieh-Klenow type of distortions, and managers' private benefits.

Each of these mechanisms can be accommodated by an extended version of our baseline model. As Appendix A.8.1, A.8.2 and A.8.3 shows, it turns out that, after taking log-quadratic approximation of the managers' payoff functions, the Markov Perfect Equilibrium under these frictions take the same form as in Part 1 of Lemma 1. This means that the belief dynamics in Lemma 2 and the MPBE characterization in Lemma 3 remain true. These frictions affect investment and hiring by changing the policy matrix F_ι and F_h, and as long as the primitives of these frictions are common knowledge within the firm, the structure of investment forecast error given in Proposition 1 still holds. Therefore, the sign pattern in Part 1 of Proposition 2 holds for all these alternative mechanisms: without communication frictions, we must have $\tilde{\beta}_h = \tilde{\beta}_\iota = 0$. In other words, these frictions cannot reproduce nonzero regression coefficients in (25) and (26) by themselves.

Behavioral Frictions The behavioral macroeconomics literature has well established that behavioral frictions in the expectation formation process makes forecast error predictable by past variables. Here, I consider replacing the rational expectation operator $\mathbb{E}_{it}$ by a "behavioral" expectation operator $\mathbb{F}_{it}$ with an ad-hoc belief updating rule

$$\mathbb{F}_{it}[a_{it}] = \rho\lambda\mathbb{F}_{i,t-1}[a_{i,t-1}] + G(s_{it} - \rho\lambda\mathbb{F}_{i,t-1}[a_{i,t-1}]) \tag{29}$$

where $s_{it} = a_{it} + \varepsilon_{it}$ is a noisy signal of productivity. Following Kohlhas and Walther (2021), I allow the Kalman gain $G > 0$ to deviate from the Bayesian Kalman gain, and the parameter $\lambda \in [0, 1/\rho]$ to be different from unity so that it captures over- and under-extrapolation, i.e. the perceived persistence of productivity shocks can be higher ($\lambda > 1$) or lower ($\lambda < 1$) than the actual persistence. This behavioral expectation nests a number of behavioral distortions considered in the literature, such as diagnostic expectation (Bordalo et al. (2022)) and belief extrapolation (Barberis et al. (2015); Barrero (2022)).

Under belief updating rule (29), the investment forecast error can be written as

$$\iota_{it} - \mathbb{F}_{i,t-1}[\iota_{it}] = F_\iota^a \rho G(a_{i,t-1} - \mathbb{F}_{i,t-1}[a_{i,t-1}]) + F_\iota^a G\rho(1-\lambda)\mathbb{F}_{i,t-1}[a_{i,t-1}] + \nu_{it} \tag{30}$$

The following proposition characterizes the sign pattern of $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ for non-Bayesian Kalman gain G and extrapolation parameter λ.

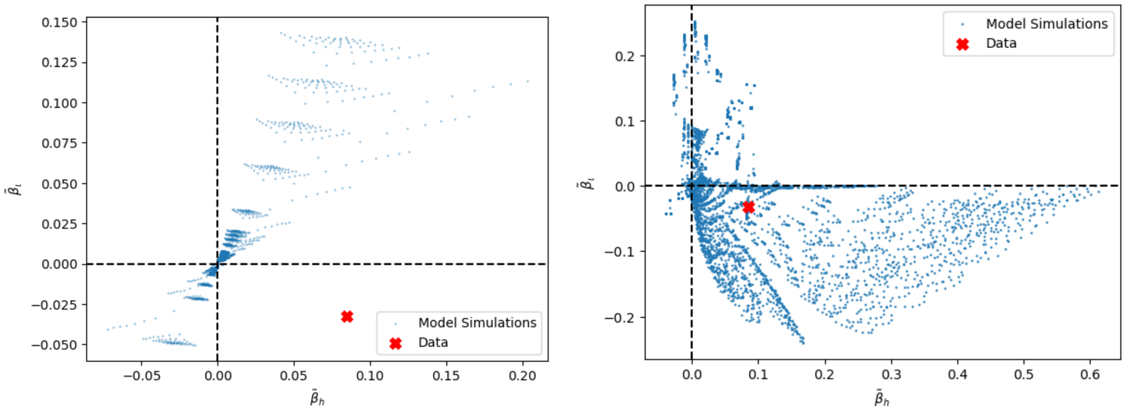
Proposition 3. *The regression coefficients $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ have the following properties if both capital and labor managers observe the same noisy signal s_{it} and form beliefs with the behavioral expectation $\mathbb{F}_{it}$:*

1. *If $\lambda = 1$, then $\tilde{\beta}_h = \tilde{\beta}_\iota = 0$ for any Kalman gain G, signal noise σ_ε and convex adjustment cost parameters (ξ_k, ξ_n);*
2. *Without adjustment cost ($\xi_k = \xi_n = 0$) nor communication friction, the signs of $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ are always the same as the sign of $1-\lambda$ under some regularity condition (A49) in Appendix A.6.*

Proof. See Appendix A.6. □

The proof and intuition of Proposition 3 are straightforward. First, from equation (30), we can see that the only reason why the regression coefficients $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ are nonzero under belief $\mathbb{F}_{it}$ is the second term on the right hand side of (30), which captures the gap between

Figure 1: $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ with Adjustment Cost, Common Noisy Info & Extrapolation



Note: In the left panel, blue dots are plotted from 4,096 model simulations with parameters $(\xi_k, \xi_n, \sigma_{\epsilon,p}, \lambda) \in [0,1] \times [0,1] \times [0.1, 5] \times [0.8, 1.07]$, equally spaced on each dimension. In each simulation, I simulate a panel of 1,000 firms for 500 periods and run the regressions (25) and (26). In the right panel, blue dots are plotted from 4.096 model simulations with parameters $(\sigma_{\epsilon,p}, \sigma_{\epsilon,k}, \sigma_{\epsilon,n}, \lambda) \in [0.1, 5] \times [0.1, 5] \times [0.1, 5] \times [0.8, 1.07]$, with adjustment cost parameters $\xi_k = \xi_n = 0.1$.

the perceived and actual persistence of productivity. If we mute this extrapolation channel by setting $\lambda = 1$, the residualized investment forecast error should not be predicted by any $t-1$ variables, including lagged hiring and investment. The non-Bayesian Kalman gain G plays no role in this case.

When beliefs are extrapolative, the regression coefficients $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ can be nonzero. However, extrapolative beliefs tend to give us a same-sign pattern for $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ rather than an opposite-sign pattern as under communication frictions. Under perfect communication, both the investment $\iota_{i,t-1}$ and hiring $h_{i,t-1}$ are known by the capital manager in period $t-1$; additionally, since capital and labor are complements, the past investment and past hiring convey similar message about the productivity to the capital manager by themselves. As a result, the belief extrapolation of the capital manager, which generates the predictability by past investment and hiring, will bias the capital manager's interpretation of investment and hiring towards the same direction.

The analytical result in Part 2 of Proposition 3 assumes no adjustment costs. With adjustment frictions, the regression coefficients can no longer be solved analytically. The left panel of Figure 1 plots model predictions on $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ by simulating the model with adjustment cost, common noisy information and extrapolative beliefs. We can see that all of the 4,096 simulations lie in the first and third quadrant, with $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ having the same signs. This reaffirms the point that opposite signs in $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ are unlikely to be reconciled with models under perfect communication. For comparison, the right panel of Figure 1 evokes communication friction on top of extrapolative belief $\mathbb{F}_{it}$ and plots the model-implied regression coefficients $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$. We can see that most of the simulated coefficients $(\tilde{\beta}_h, \tilde{\beta}_\iota)$ fall on the fourth quadrant with $\tilde{\beta}_h > 0$ and $\tilde{\beta}_\iota < 0$, the quadrant where the US data lies at. It's much easier for models with communication frictions to match the opposite-sign pattern.

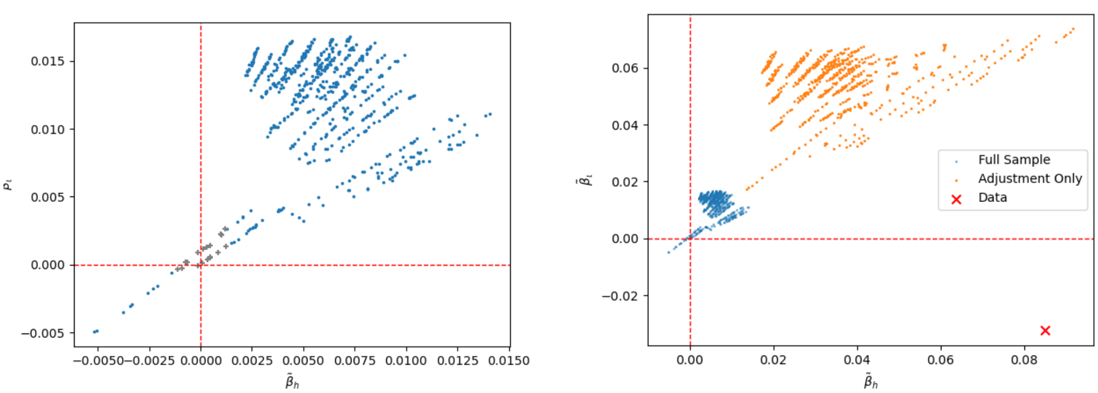
Non-convex Adjustment Cost Besides behavioral distortions, non-convex adjustment costs in capital and labor can be another reason why $\tilde{\beta}_h$ or $\tilde{\beta}_\iota$ is nonzero under common information: since fixed adjustment cost creates lumpiness in adjustment, each nonzero investment picks up information since last adjustment, making it possible for past hiring or investment to predict future forecast errors in investment. Can non-convex cost indeed generate nonzero regression coefficients and reproduce the opposite-sign pattern without the help of communication friction?

I answer these questions numerically by augmenting the baseline model with fixed adjustment costs on both capital and labor: if the capital manager makes a nonzero investment, he needs to pay a fixed fraction χ_k of total revenue as fixed costs; similarly, if the labor manager makes a nonzero hiring, she needs to pay a fixed fraction χ_n of total revenue as fixed costs. This setup makes fixed cost scales with firm size, as in Cooper and Haltiwanger (2006) and Bloom (2009). I solve the model numerically and simulate a panel of 1,000 firms for 500 periods to obtain the model-implied regression coefficients, with 625 combinations of convex and non-convex adjustment cost parameters. The results are scatter-plotted in Figure 2.

The numerical simulations give us two messages. First, the magnitude of $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ is small under realistic inaction rates. In the simulations in Figure 2, the inaction rate of capital and labor adjustments spans from 0 to 50 percent (in comparison, the inaction rate in the data is about 10 percent for both capital and labor). However, the regression coefficients are hardly above 0.01, falling short of the magnitude in the data. Figure 3 explains the reason why the regression coefficients are not sensitive to non-convex costs. As we increase the fixed adjustment cost in capital, two counteracting forces impede the $\tilde{\beta}_h$ or $\tilde{\beta}_\iota$ from increasing. On the one hand, under larger fixed cost, capital investment becomes more lumpy, and each adjustment picks up more past information, making the investment forecast error more predictable by past hiring or investment, as shown by the orange line in Figure 3. On the other hand, the inaction rate increases as the fixed cost increases, and regression coefficients for these non-adjustment are strongly negative because a large realization of investment or hiring in period $t-1$ tends to increase the investment forecast, making the investment forecast error more negative when $\iota_{it} = 0$ (inaction). This counteracting force, as shown by the green line in Figure 3, drags down the $\tilde{\beta}_h$ or $\tilde{\beta}_\iota$ in the full sample. As a result, the regression coefficients are close to 0 and insensitive to the non-convex cost parameters.

Second, it is unlikely for the nonconvex cost to present opposite signs for $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ under common information. We can see from Figure 2 that the regression coefficients $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ are mostly in the first and third quadrant. None of the coefficients lie in the fourth quadrant, where the data is at. The same pattern is true for the regression coefficients conditional on adjustment in the right panel of Figure 2: all the $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ are positive. The intuition for this numerical result is similar: without communication frictions, observing the intensive and extensive margin of investment and hiring convey similar messages about productivity to the capital manager. Although the fixed adjustment costs generate inaction and lumpy adjustment, they are not able to generate episodes with investment and hiring going in opposite directions. Communication friction, on the other hand, can generate such patterns much more easily.

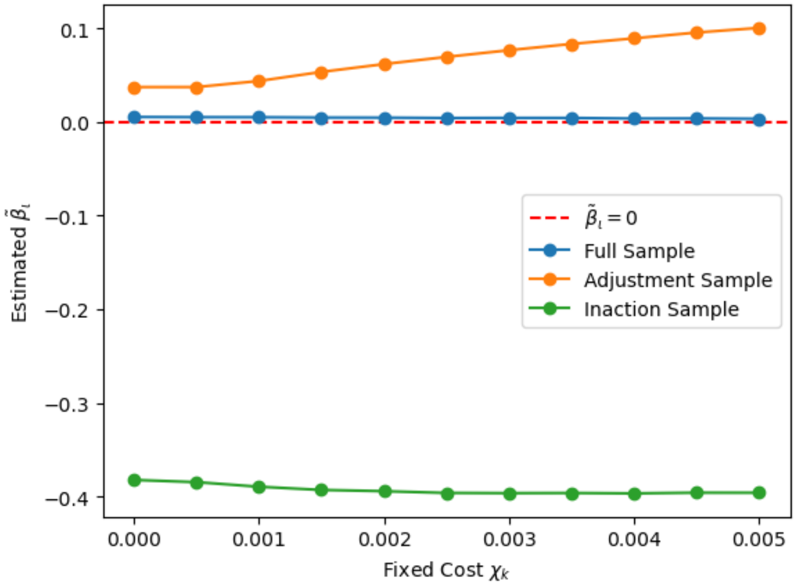
Figure 2: $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ under Non-convex Adjustment Cost and Common Information



Note: In the left panel, blue dots are plotted from 625 model simulations with parameters $(\xi_k, \xi_n, \chi_k, \chi_n) \in [0,2] \times [0,2] \times [0,0.005] \times [0,0.005]$, equally spaced on each dimension, with $\sigma_{\epsilon,p} = 0.3$. In each simulation, I simulate a panel of 1,000 firms for 500 periods and run the regressions (25) and (26). The inaction rate of these simulations spans from 0 to 50 percent for both capital and labor. The dots in gray are those with p-value greater than 0.1 in the simulated panel. In the right panel, the orange dots plot the regression coefficients conditional on adjustment under the same simulation panel.

Summary In order to identify communication friction, I propose to run both regression (25) and regression (26) and test if the estimated coefficients $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ have the same or opposite sign. An opposite sign of predictability by past actions is difficult to be reconciled by models without communication friction and indicative of the existence of imperfect

Figure 3: Reason for Small $\tilde{\beta}_h$ under Non-convex Cost



communication between capital and labor decisions.

3.4 Application to Data

I now apply the test (25) and (26) to a merged firm-level expectation-outcome dataset for US public firms.

Dataset For the expectation data, I use the IBES Guidance dataset, which contains all the guidance activities (i.e. the firm's public announcement on its own expectations about sales, capital expenditure, earnings per share, among many others, for the current fiscal year) of US public firms since 1992. These guidances can be regarded as the self-reported expectation of the firm on its own financial performances. Since all measurements in the guidance are financial indicators, I assume that the expectations posted in IBES Guidance mainly reflect the capital manager's view about the firm's performance.

According to (25), the firm-level expectation that we are interested in is (1) the forecast of investment $\mathbb{E}^k_{i,t-1}[\iota_{it}]$ and (2) the nowcast of sales in period $t-1$, $\mathbb{E}^k_{i,t-1}[y_{i,t-1}]$. These two metrics can be proxied by the sales guidance and capital expenditure[7] guidance, both of which

[7]Here, I regard capital expenditure (CPX) as a proxy for investment, although it is not exactly the same as ι_{it} in the model.

are widely covered in the IBES dataset as they are the most frequently reported guidances by the US public firms. However, since guidance activities are self-reported and can happen in any time of the fiscal year, it is difficult to get a notion for the time horizon of the forecast from the announcement date. Here, I restrict my attention to the firms that have multiple guidance activities within the same year, and in that given year, I take the first posted capital expenditure guidance as a proxy for the one-period-ahead forecast of investment $\mathbb{E}^k_{i,t-1}[\iota_{it}]$, and I take the last posted sales guidance as a proxy for the nowcast of sales $\mathbb{E}^k_{i,t-1}[y_{i,t-1}]$. I then merge the IBES Guidance dataset to CRSP/Compustat Fundamental Annual and get a firm-level dataset with both expectations and realized outcomes. The details of data cleaning and sample construction are laid out in Appendix B.1.

The resulting IBES-Compustat merged dataset spans over 2000 and 2024 with 8,450 firm-year observations. Table B1 shows that the IBES-Compustat merged dataset is a sample for gigantic firms in the US economy: on average, firms in the merged sample is about two times larger than those in the CRSP-Compustat dataset in terms of sales and employment. Although the merged sample covers only 11% of total observations in the CRSP-Compustat Fundamental Annual Dataset between 2000 and 2024, the total sales of firms in the merged IBES-Compustat sample take up about 20% of US GDP in a sample year.

For each firm-year observation in the IBES-Compustat merged sample, I take the realized sales and capital expenditure from Compustat and impute the forecast/nowcast error as the arc-percent difference between the predicted value and the actual value, as in Davis and Haltiwanger (1992) and Bloom, Codreanu, and Fletcher (2025):

$$\text{CapEX Forecast Error}_{it} = \frac{\text{CapEX}_{it} - \mathbb{E}^k_{i,t-1}[\text{CapEX}_{it}]}{1/2(\text{CapEX}_{it} + \mathbb{E}^k_{i,t-1}[\text{CapEX}_{it}])}$$

and

$$\text{Sales Nowcast Error}_{it} = \frac{\text{Sales}_{it} - \mathbb{E}^k_{i,t}[\text{Sales}_{it}]}{1/2(\text{Sales}_{it} + \mathbb{E}^k_{i,t}[\text{Sales}_{it}])}$$

The arc-difference definition of forecast/nowcast errors has two advantages over the log difference. First, it is bounded between $[-2, 2]$, so that outliers in the realized values and expectations would not affect the forecast/nowcast error and the regression results by much. Additionally, it avoids the problems of taking logs when the actual values or expectations are close to 0. This may not be a big issue for the large-firm samples like IBES or Compustat, and I verify that the regression results are robust to alternative definitions of forecast/nowcast errors (e.g. the log difference between predicted values and actual values).

Regression Results Table 1 tabulates the regression result for the joint test of (25) and (26). The first column shows the result for regression (25). We can see that $\tilde{\beta}_h$ is positive and statistically significant, which implies that the investment expectation of US public firms systematically under-react to their own past hiring. The second column shows the result for regression (26). We can see that $\tilde{\beta}_\iota$ is negative and statistically significant, which implies that the investment expectation of US public firms systematically over-react to their own past investment. This is consistent with the findings in Barrero (2022), which documents a similar pattern of belief overreaction for sales expectations of US firms. The third column considers including both lagged hiring and lagged investment in the regression, and these multivariate estimates are similar to the separate estimates in the first two columns. It shows that the correlation between past investment and past hiring does not bias the separate regression results in the first two columns. The opposite-sign pattern of $\tilde{\beta}_h > 0$ and $\tilde{\beta}_\iota < 0$ is consistent with the predictions in Proposition 2 and can only be reconciled by models with communication frictions.

Table 1: Predictability of Investment Forecast Error by Past Hiring and Investment

	Dependent variable:		
	$\iota_t - \mathbb{E}^k_{t-1}[\iota_t]$		
	(1)	(2)	(3)
h_{t-1}	0.096^{***}		0.104^{***}
	(0.021)		(0.021)
ι_{t-1}		-0.033^{***}	-0.037^{***}
		(0.008)	(0.008)
$y_{t-1} - \hat{y}^k_{t-1}$	0.089^{***}	0.113^{***}	0.095^{***}
	(0.031)	(0.031)	(0.031)
Firm Fixed Effect	Yes	Yes	Yes
Sector-Year Fixed Effect	Yes	Yes	Yes
Observations	8,455	8,455	8,455
R^2	0.562	0.562	0.563
Note:	*p<0.1; **p<0.05; ***p<0.01		

I now address two robustness concerns. The first one is regarding the timing of expectations. Since firms post their guidance in a discrete manner, the first capital expenditure guidance of a year may be posted in the second, third or even the fourth quarter and therefore may not reflect the notion of "forecast". Similarly, the last sales expenditure of a year may be posted way earlier than the fourth quarter of the year and does not capture the notion of "nowcast". To mitigate the concern, I re-run the regressions in Table 1 on a subsample that contains only the first-quarter forecast of capital expenditure in year t and the last-quarter forecast of sales in year $t-1$. I tabulates the result in Table 2. We can see that the main takeaway remains the same as in the full sample: the estimates of $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ are similar to those in the full sample, statistically significant, and have the same sign pattern as in Table 1. This partially address the concern that the results are driven by not insufficient controls of fundamental uncertainty or invalid proxies. That being said, I shall mention that even in this restricted sample, the timing in data is not entirely the same as in the model: the Q4-expectation of sales in year $t-1$ and the Q1-expectation of forecasts in year t are treated as if they are conditional on the same information set $\mathcal{I}^k_{i,t-1}$, which implicitly assumes that there is no new information flows between the Q4 of year $t-1$ and Q1 of year t. This inconsistency will be better addressed using a firm sample that elicit forecast and nowcast/backcast expectations under a fixed time frame.

Table 2: Predictability under Strict Timing

	Dependent variable:		
	$\iota_t - \mathbb{E}^k_{t-1}[\iota_t]$		
	(1)	(2)	(3)
h_{t-1}	0.083***		0.090***
	(0.031)		(0.031)
ι_{t-1}		−0.035***	−0.038***
		(0.012)	(0.012)
$y_{t-1} - \hat{y}^k_{t-1}$	0.193***	0.214***	0.200***
	(0.057)	(0.057)	(0.057)
Firm Fixed Effect	Yes	Yes	Yes
Sector-Year Fixed Effect	Yes	Yes	Yes
Observations	4,327	4,327	4,327
R^2	0.656	0.656	0.657
Note:	*p<0.1; **p<0.05; ***p<0.01		

Another concern is whether the results in Table 1 are driven by outliers. To address the concern, I trim the 1 percent tails for investment forecast error and sales nowcast error in the full sample and rerun the regression. The results are reported in Table 3. We can see that the estimated coefficients $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ are both statistically significant at 1% level and have the same sign pattern as in Table 1. The magnitude of $\tilde{\beta}_\iota$ in the trimmed sample is almost identical to the estimates in the full sample, but the magnitude of $\tilde{\beta}_h$ is about 25% smaller than the full-sample estimates. Since the dropped observations are mostly from very large companies, this is likely indicating that communication friction is weaker for smaller companies.

Table 3: Predictability with Trimmed Sample

	Dependent variable:		
	$\iota_t - \mathbb{E}^k_{t-1}[\iota_t]$		
	(1)	(2)	(3)
h_{t-1}	0.067***		0.076***
	(0.021)		(0.021)
ι_{t-1}		−0.034***	−0.036***
		(0.007)	(0.007)
$y_{t-1} - \hat{y}^k_{t-1}$	0.241***	0.287***	0.246***
	(0.063)	(0.062)	(0.063)
Firm Fixed Effect	Yes	Yes	Yes
Sector-Year Fixed Effect	Yes	Yes	Yes
Observations	8,099	8,099	8,099
R^2	0.517	0.518	0.519
Note:			*p<0.1; **p<0.05; ***p<0.01

In summary, in this section I describe the strategy of identifying communication friction from the expectation data. The US public firm's managerial expectation data shows that the investment forecast simultaneously over-reacts to the firm's past investment and under-reacts to the firm's past hiring. I show that this sign pattern cannot be easily reconciled by models without communication friction and is consistent with the implications of communication friction.

4 Quantitative Analysis

In this section, I quantify the aggregate effect of communication friction. I will first extend the model in Section 2 to incorporate other well-documented distortions and frictions in the firm dynamics literature. Then, I will discuss the calibration strategy that allows me to identify the parameters for each mechanism in the extended model. After estimating the extended model, I will use the calibrated parameters to investigate the role communication friction plays in accounting for the patterns in the data.

4.1 Calibration

An Extended Model for Quantification To obtain a credible quantification of aggregate effect of communication friction, I extend our baseline model so that it incorporates Hsieh-Klenow type of reduced-form distortions. The firm's profit function becomes

$$\begin{aligned}\Pi_{it} = &(1-\tau_{it}^{y})R(K_{it}, N_{it}; A_{it}) - (1+\tau_{it}^{k})R_t K_{it} - W_t N_{it} \\ &- \frac{\xi_k}{2}\left(\frac{K_{it}}{K_{i,t-1}} - 1\right)^2 K_{i,t-1} - \frac{\xi_n}{2}\left(\frac{N_{it}}{N_{i,t-1}} - 1\right)^2 N_{i,t-1}\end{aligned} \tag{31}$$

where $R(K_{it}, N_{it}; A_{it})$ is the revenue function, and τ_{it}^{y} and τ_{it}^{k} are reduced-form firm-specific distortions as in Hsieh and Klenow (2009). τ_{it}^{y} captures the mechanisms that distort all factors symmetrically without changing the capital-labor ratio (such as price markup), while τ_{it}^{k} captures the capital-specific distortions (such as financial frictions) that create variations the capital-labor ratio. I assume that the distortions follow the distribution $\tau_{it}^{y} \sim N(0, \sigma_{\tau,y}^2)$ and $\tau_{it}^{k} \sim N(0, \sigma_{\tau,k}^2)$ and are i.i.d across firms and over time, uncorrelated with each other and independent of other shocks.

Calibration In total, there are 7 parameters to be calibrated in the extended model: convex adjustment costs ξ_k and ξ_n, dispersion of uncorrelated distortions $\sigma_{\tau,k}$ and $\sigma_{\tau,y}$, and the behavioral/information parameters $(\sigma_{\epsilon,p}, \sigma_{\epsilon,k}, \sigma_{\epsilon,n})$.

The extended model will be estimated by a Simulated Method of Moments (SMM) approach. To pin down the information parameters, I include the regression coefficient $\tilde{\beta}_h$ ad $\tilde{\beta}_\iota$ as targeted moments, since they jointly identify communication frictions from other mechanisms. To pin down the adjustment costs and dispersion of distortion, I follow David and

Venkateswaran (2019) and target serial correlation of investment and hiring, the variances of investment and hiring, and the investment-hiring correlation in the calibration. Since adjustments costs create an intertemporal smoothing motives for capital and labor adjustment, high adjustment costs create large serial correlation and small volatility in investment and hiring, while information frictions and distortions amplify the investment and hiring volatility. The investment-hiring correlation helps us distinguish between the mechanisms that affect inputs symmetrically (such as τ^y) and input-specific distortions (such as τ^k).

The Simulated Method of Moments estimation is done in the following steps:

1. For each parameter $\Theta = (\xi_k, \xi_n, \sigma_p, \sigma_{\epsilon,k}, \sigma_{\epsilon,n}, \sigma_{\tau,k}, \sigma_{\tau,y})$, solve the MPBE according to Lemma 1, 2 and 3;
2. Simulate a panel of 1,000 firms for 500 periods and compute the simulated moments $m(\Theta)$ from the panel;
3. Estimate the parameter by choosing the one that minimizes the sum of quadratic deviations of simulated moments from the data moments m:

$$\hat{\Theta} = \arg\min_{\Theta}(m(\Theta) - m)'(m(\Theta) - m)$$

The targeted moments and estimated parameter values are summarized in Table 4.

Table 4: Estimated Parameter Values and Targets

Description		Value	Source/Target
Elasticity of substitution	ε	4	Standard Value
Capital share	α	0.47	Cost share in IBES/Compustat
Persistence of Productivity	ρ	0.94	Estimates of $a_{it} = \rho a_{i,t-1} + \mu_{it}$.
Std.dev. of Productivity shock	σ_μ	0.24	
Capital convex adj. cost	ξ_k	1.84	$\text{corr}(\iota, \iota_{-1})$
Labor convex adj. cost	ξ_n	2.13	$\text{corr}(h, h_{-1})$
Noise on the common signal	$\sigma_{\epsilon,p}$	3.46	Reg. Coef. $\tilde{\beta}_\iota$ from (26)
Noise on K manager's signal	$\sigma_{\epsilon,k}$	2.53	Reg. Coef. $\tilde{\beta}_h$ from (25)
Noise on N manager's signal	$\sigma_{\epsilon,n}$	1.43	$\text{corr}(\iota, h)$
Dispersion of Capital Distortion τ^k	$\sigma_{\tau,k}$	1.09	$\text{var}(\iota)$
Dispersion of Revenue Distortion τ^y	$\sigma_{\tau,y}$	0.37	$\text{var}(h)$

Data The data moments are computed from a merged IBES/Compustat sample for US public firms between 2000 to 2024. From the Compustat sample, I take the book value of property, plant and equipment (PPE) as the measure for capital stock. Investment rate ι and hiring rate h are obtained by first-differencing the logged capital and employment series respectively. Value-added is constructed by sales minus the material expenditure.[8] I then construct the total revenue factor productivity (TFPR) or sales Solow residual as the firm fundamental a_{it} by

$$a_{it} = (1-1/\epsilon)^{-1} \left[\log(\text{Value-added}) - \alpha(1-1/\epsilon)\log(\text{PPE}) - (1-\alpha)(1-1/\epsilon)\log(\text{Employment})\right]$$

where the capital share α is chosen to match the cost share of labor in the sample, and the elasticity of substitution $\varepsilon = 4$ is taken from the lower bound of the literature (Hsieh and Klenow (2009)). As standard in the literature, the measure for capital and labor misallocation is the dispersion of the average/marginal revenue product of capital and labor (MRPK or MRPN). I define MRPK as the difference between logged value-added and logged capital and MRPN as the difference between logged value-added and logged employment. Following David and Venkateswaran (2019), I regress the MPRK, MRPN, investment, hiring and firm fundamental series on a sector-year fixed effect and keep the residual to isolate the firm-specific variation in each series.[9] I then merge the IBES Guidance, which contains firms' self-reported expectations, to the Compustat sample. Same as before, I use the first capital expenditure guidance of the year as "investment forecasts" and the last sales guidance of the year as "sales nowcast". I trim 1% tail of each variable in the merged sample, and then keep the observations that have investment, lagged investment, lagged hiring, lagged sales, capital expenditure forecast and lagged sales nowcast. The resulting sample has 6,492 firm-year observations, which I use to compute the data moments in the calibration. I relegate further details of sample construction and production function estimation to Appendix B.2.

4.2 Results

Communication Friction and Regression Coefficients Table 5 shows that our model can match the data moments almost exactly. In particular, the model-implied regression coefficients $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ match the data moments with a noisy firm-level common signal and

[8] Material expenditure is defined as Cost of Goods Sold (COGS) + Selling, General & Administrative Expenses (XSGA) - Depreciation (DP) - Wage Bill (XLR), following the approach by Keller and Yeaple (2009) and Flynn and Sastry (2020). Since wage bill is missing for about 90% of the firm, I use the two-digit industry-level wages in the Census Bureau County Business Patterns dataset to impute the wage bill for the firm. See Appendix B.2 for details.

[9] Here, sector or industry is defined as 2-digit NAICS code for non-manufacturing industries and 3-digit NAICS code for manufacturing industries.

more precise manager-level signals. As a comparison, I re-calibrate a simplified version of the model to the same set of targets, muting communication frictions by setting manager's private information to be extremely noisy ($\sigma_{\epsilon,k}, \sigma_{\epsilon,n} \to \infty$), and allowing for extrapolative expectation as in (29). The results are tabulated on the third column of Table 5. We can see that the model without communication friction is not able to match the sign pattern of the two regression coefficients. Under the common-information calibration, the model-implied predictability by lag hiring and lag investment are both zero. Our baseline specification with communication friction can easily match the opposite-sign pattern in the data. This is consistent with the takeaway in Proposition 3 and Figure 1.

Table 5: Targeted Moments: Data vs Model

Moment	Data	Model (Baseline)	Model (Common Info & Behavioral $\mathbb{F}_{it}$)
corr(ι, ι_{-1})	0.294	0.293	0.293
corr(h, h_{-1})	0.178	0.178	0.178
corr(ι, h)	0.526	0.526	0.653
$\tilde{\beta}_h$	**0.086**	**0.086**	**0.000**
$\tilde{\beta}_\iota$	**-0.034**	**-0.034**	**0.000**
var(ι)	0.018	0.013	0.011
var(h)	0.020	0.023	0.022

Sources of Learning The estimated parameters shed light on the learning patterns of the capital and labor managers. Table 6 shows how much the managers learn from multiple information resources. Repeatedly learning from noisy signals leads to a substantial reduction in uncertainty for both managers: posterior uncertainty is less than 50 percent of prior uncertainty for both the capital and labor manager. In this baseline calibration, managers' private signals play an important role in decision making. Both managers put a non-trivial weight on their private signals, so that the model matches the sign patterns of regression coefficients in the expectation test and the low investment-hiring correlation in the data. These private signals generate different posterior beliefs at the time investment and hiring decisions are made and breaks the perfect correlation between capital and labor manager's beliefs under perfect communication. The fact that productivity is never fully revealed is important because it keeps the prior beliefs imperfect over time, so within-period private information continues to matter for decision-making.

Table 6: Learning Patterns of the Managers

	Uncertainty		Source of Learning		
	Posterior	Posterior/Prior	Public	Private	Prior
Capital Manager	0.193	47.4%	1.60%	3.02%	95.38%
Labor Manager	0.181	44.5%	1.50%	8.91%	89.59%

Sources of Low Investment-hiring Correlation Low investment-hiring correlation is an important feature in the US firm data. Which mechanisms are responsible for it, and which one is the most prominent? In Table 7, I report the contribution of each friction to the investment-hiring correlation. The numbers in the table are computed under the assumption that only one friction of interest is active (i.e. the relevant parameters are fixed at the estimated level) while all other frictions are muted (i.e. the relevant parameters are set to the frictionless level).

From Table 7, we can see that the low investment-hiring correlation is primarily driven by communication friction. Each of other frictions generate almost perfect correlation in investment and hiring. Additionally, the information friction is a much more powerful force in driving investment-hiring correlation down: with information friction only, we can even generate a negative correlation between investment and hiring. This is not entirely absurd: Hawkins, Michaels, and Oh (2015) documents that firms and plants have episodes when capital and labor adjustments are in opposite directions. The leading explanation is that the complementarity between capital and labor varies with tasks (Acemoglu and Restrepo (2019)). Our result shows that communication friction provides a complementary explanation.

Investment-hiring correlation highlights a crucial difference between firm-level public information and manager-level private information. We can see that if we mute communication friction and only allow for firm-level public information, the investment-hiring correlation is equal to 1. Manager-level private signals, on the other hand, lower the correlation significantly. This can be seen from the expressions of capital and labor choices. Without other frictions, capital and labor are both proportional to the managers' posterior beliefs about productivity, i.e. $k_{it} \propto \mathbb{E}^k_{it}[a_{it}]$ and $n_{it} \propto \mathbb{E}^n_{it}[a_{it}]$. Absent communication friction, managers share the same information, so that $\mathbb{E}^k_{it} = \mathbb{E}^n_{it} = \mathbb{E}_{it}$, and firm-level common noisy information attenuates the responsiveness of both inputs by the same proportion, without breaking the perfect correlation between them. The existence of private signals breaks this pattern as the managers' expectations are no longer perfectly correlated in general. Moreover, in my model, since the true productivity is never fully revealed to the managers, the disagreement between managers on productivity is preserved over time, and hence the prior beliefs of the managers are different from each other. This largely amplifies the effect of imperfect communication on the capital-labor correlation.

The distinction between firm-level public information and manager-level private information is akin to the difference between revenue distortion τ^y and capital distortion τ^k. We can see from the table that uncorrelated capital distortion τ^k lowers the investment-hiring correlation, while the the uncorrelated revenue distortion τ^y doesn't. Common, noisy information acts like the revenue distortion τ^y as they both affect all the inputs symmetrically, while the private information introduced by communication friction acts like the capital-specific distortion τ^k, as they both affect different inputs disproportionately and generate variations in capital-labor ratio.

Table 7: Sources of Capital-labor Disconnect: Baseline Model

	corr(ι, h)
Convex Costs	0.997
- Capital Adjustment Cost ξ_k	0.875
- Labor Adjustment Cost ξ_n	0.912
Information frictions	-0.006
- Public Signal $\sigma_{\epsilon,p}$	1.000
- Capital Manager's Private Signal $\sigma_{\epsilon,k}$	-0.034
- Labor Manager's Private Signal $\sigma_{\epsilon,n}$	-0.033
Transitory, Uncorrelated Distortion τ	0.978
- Transitory, Uncorrelated Capital Distortion $\sigma_{\tau,k}$	0.976
- Transitory, Uncorrelated Revenue Distortion $\sigma_{\tau,y}$	1.000
All mechanisms	0.526
Data	0.526

Sources of Variations in the Capital-labor Ratio All the three mechanisms (adjustment costs, communication friction, and capital-specific distortion) can qualitatively distort and generate dispersion in capital-labor ratio within and across firms. Is communication friction an important source of variations in capital-labor ratio quantitatively?

To answer this question, it is useful to follow Kaymak and Schott (2023) and write the first-order conditions (12) and (13) in the Markov Perfect Bayesian Equilibrium as

$$-\tau_{it}^{y}+\hat{a}_{it}^{k}(1-1/\epsilon)+(\hat{\alpha}_1-1)k_{it}+\hat{\alpha}_2\hat{n}_{it}^{k}+\omega_{it}^{k}=\tau_{it}^{k} \tag{32}$$

$$-\tau_{it}^{y}+\hat{a}_{it}^{n}(1-1/\epsilon)+\hat{\alpha}_1\hat{k}_{it}^{n}+(\hat{\alpha}_2-1)n_{it}+\omega_{it}^{n}=0 \tag{33}$$

where $\hat{n}_{it}^{k}\equiv\mathbb{E}_{it}^{k}[n_{it}]$ is the capital manager's perceived employment in period t, $\hat{k}_{it}^{n}$ is the labor manager's perceived capital stock in period t, and ω_{it}^{k} and ω_{it}^{n} are the marginal continuation value net of marginal adjustment cost for capital and labor. Under the linear-quadratic setup, these ω's are linear functions of capital, labor and productivity expectation, and they can be interpreted as an endogenous wedge generated by adjustment costs: absent adjustment costs, $\omega^k=\omega^n=0$, and the FOCs degenerate to the static ones in Hsieh and Klenow (2009). I relegate the details to Appendix A.7.

Expressing the FOCs in this static-like form significantly simplifies the decomposition of mechanisms in firm dynamics. For simplicity, from now on I drop the time and firm subscript. The following proposition analytically characterizes the contributions of adjustment costs, information frictions and reduced-form distortions to the variations in capital-labor ratio.

Proposition 4. *The reformulated FOCs (32) and (33) implies that capital-labor ratio $k-n$ can be written as*

$$k-n=-\tau^k+\left[(\epsilon-1)+\hat{\alpha}_2\epsilon A^{(n,k)}+\hat{\alpha}_1\epsilon A^{(k,n)}\right](\hat{a}^k-\hat{a}^n)+\omega^k-\omega^n \tag{34}$$

and the variation in capital-labor ratio can be decomposed as

$$var(k-n)=\underbrace{\sigma_{\tau,k}^2}_{\text{Capital Distortion}}+\underbrace{\sigma_{\omega,k}^2+\sigma_{\omega,n}^2-2cov(\omega^k,\omega^n)}_{\text{Direct Effect of Adjustment Cost}}+\underbrace{2cov(\tau^k,\omega^n)-2cov(\tau^k,\omega^k)}_{\text{Adj Cost}\times\text{Distortion}} \tag{35}$$

$$+\underbrace{\left[(\epsilon-1)+\hat{\alpha}_2\epsilon A^{(n,k)}+\hat{\alpha}_1\epsilon A^{(k,n)}\right]^2\hat{\Sigma}^{(k,n)}}_{\text{Direct Effect of Imperfect Communication}} \tag{36}$$

$$+\underbrace{2\left[(\epsilon-1)+\hat{\alpha}_2\epsilon A^{(n,k)}+\hat{\alpha}_1\epsilon A^{(k,n)}\right]cov(\hat{a}^k-\hat{a}^n,\omega^k-\omega^n)}_{\text{Adj Cost}\times\text{Imperfect Communication}} \tag{37}$$

where $\hat{\Sigma}^(k,n)\equiv\mathbb{E}[(\hat{a}^k-\hat{a}^n)^2]=\hat{\Sigma}^k+\hat{\Sigma}^n-2\tilde{\Sigma}^{(k,n)}$ as in Lemma 3, and $A^{(k,n)}$, $A^{(n,k)}$ are constants.

Proof. See Appendix A.7. □

Equation (34) shows that capital-labor ratio is a linear combination of capital-specific distortion τ^k, the difference between capital and labor manager's posterior $\hat{a}^k - \hat{a}^n$, and adjustment-cost wedges ω^k, ω^n. We can see that apart from adjustment costs and capital distortions, communication friction contributes directly to the capital-labor ratio, by breaking the consensus of capital and labor manager on productivity, i.e. $\hat{a}^k \neq \hat{a}^n$. Without communication frictions, capital and labor manager share the common information and hold the same belief, and adding noise to common information will affect the capital-labor ratio only through the adjustment-cost wedges ω^k and ω^n. It is also worth noting that the second and third term in the coefficient of $\hat{a}^k - \hat{a}^n$ come from the fact that, under communication friction, the capital and labor manager do not know the other manager's adjustment-cost wedge perfectly. This makes way for the adjustment cost parameters to directly affect the contribution of this channel to capital-labor ratio (by changing the constants $A^{(k,n)}$ and $A^{(n,k)}$).

The expression from (35) to (37) allows us to quantitatively decompose the source of capital-labor ratio. Line (35) collects the contributions of adjustment costs and reduced-form distortions. Line (36) summarizes the direct effect of imperfect communication on capital-labor ratio dispersion, and line (37) collects the interaction between adjustment cost and communication frictions. Our parameterization in Table 4 allows us to compute each of the components using analytical or simulated moments:

$$\underbrace{0.046}_{var(k-n)} = \underbrace{1.204}_{\text{Distortion}} + \underbrace{1.117}_{\text{Adj Cost}} + \underbrace{-2.273}_{\text{Adj Cost} \times \text{Distortion}} + \underbrace{0.027}_{\text{Communication Friction}} + \underbrace{-0.027}_{\text{Communication Friction} \times \text{Adj Cost}} \tag{38}$$

Interestingly, under the current parameterization, communication friction is not a major source of dispersion in capital-labor ratio. Although the direction effect of imperfect communication explains more than half of the capital-labor ratio dispersion, its interaction with adjustment frictions completely offsets the direct effect. Moreover, the interaction between adjustment cost and communication friction is not sensitive to the change in adjustment cost parameters. In Table 8, I consider down-scaling the adjustment cost paramters to 60% and to 20% of their estimated values and report the resulting quantification in the second and third column. We can see that as adjustment costs decrease, the net contribution of communication friction to the capital-labor ratio variability goes up from 0 to around 20% of total capital-labor ratio dispersion. And this is fully driven by the fact that lower adjustment costs amplify the direct effect of communication friction, as the interaction term between communication friction and adjustment cost remains unchanged at around -0.027. This interaction term largely impedes the communication friction to play a quantitatively significant role in the dispersion of capital-labor ratio.

Table 8: Communication Friction × Adjustment Cost, Alternative Parameters

Adj. Cost Parameters	$\xi_k = 1.84$ $\xi_n = 2.13$	$\xi_k = 1.10$ $\xi_n = 1.28$	$\xi_k = 0.37$ $\xi_n = 0.43$
var$(k - n)$	0.046	0.060	0.130
Communication Friction	**-0.001**	**0.004**	**0.024**
Direct	0.027	0.032	0.051
Interaction with Adj Cost	-0.027	-0.028	-0.027

Besides the interaction with adjustment costs, the contribution of communication frictions to the dispersion in capital-labor ratio depends on the degree of strategic complementarity between capital and labor, which is captured by the cross-partial derivative of profit function $\alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y}$. The capital and labor presents stronger strategic complementarity as the elasticity of substitution ϵ gets larger or the capital share of income α gets closer to 0.5. Table 9 shows that the contribution of communication friction to capital-labor ratio dispersion becomes larger as strategic complementarity gets stronger (as ϵ goes up). Intuitively, as the strategic complementarity between capital and labor get stronger, each manager's uncertainty about the other manager's decision will have a larger impact on the expected marginal revenue product. However, even with a large ϵ, communication frictions do not appear to be a significant source of capital-labor ratio dispersion: the total contribution is less than 10 percent when $\epsilon = 11$, the upper bound in the literature. This is again because the offsetting interaction force between communication friction and adjustment costs also gets larger as strategic complementarity gets stronger.

Table 9: Strategic Complementarity and Capital-labor Ratio

	$\epsilon = 4$	$\epsilon = 6$	$\epsilon = 9$	$\epsilon = 11$
var$(k - n)$	0.046	0.078	0.118	0.140
Communication Friction	**-0.001**	**0.003**	**0.009**	**0.012**
Direct	0.027	0.041	0.056	0.064
Interaction with Adj Cost	-0.027	-0.038	-0.047	-0.052

Allocative Efficiency Lastly, I investigate the role of communication frictions in capital and labor misallocation. Using the same method, I characterize the decomposition of MRPK and MRPN into adjustment costs, information frictions, distortions and the interaction terms in the following proposition:

Proposition 5. *The reformulated FOCs (32) and (33) implies that the dispersion in MRPK and MRPN can be decomposed as*

$$var(mrpk) = \underbrace{\sigma^2_{\tau,k} + \sigma^2_{\tau,y}}_{\text{Exo. Distortions}} + \underbrace{\sigma^2_{\omega,k}}_{\text{Direct Effect of Adj Cost}} \underbrace{-2cov(\tau^k, \omega^k) - 2cov(\tau^y, \omega^k)}_{\text{Adj Cost} \times \text{Distortion}} \tag{39}$$

$$+ \underbrace{(A^k + A^n)(A^k\hat{\Sigma}^k + A^n\hat{\Sigma}^n) - A^kA^n\hat{\Sigma}^{(k,n)}}_{\text{Direct Effect of Information Frictions}} \tag{40}$$

$$\underbrace{-2A^k cov(\hat{a}^k - a, \omega^k) - 2A^n cov(\hat{a}^n - a, \omega^k)}_{\text{Info Friction} \times \text{Adj Cost}} \tag{41}$$

and

$$var(mrpn) = \underbrace{\sigma^2_{\tau,y}}_{\text{Exo. Distortion}} + \underbrace{\sigma^2_{\omega,n}}_{\text{Adj Cost}} \underbrace{-2cov(\tau^y, \omega^n)}_{\text{Distortion} \times \text{Adj Costs}} \tag{42}$$

$$+ \underbrace{(B^k + B^n)(B^k\hat{\Sigma}^k + B^n\hat{\Sigma}^n) - B^kB^n\hat{\Sigma}^{(k,n)}}_{\text{Info Frictions}} \tag{43}$$

$$- \underbrace{2B^k cov(\hat{a}^k - a, \omega^n) - 2B^n cov(\hat{a}^n - a, \omega^n)}_{\text{Info Friction} \times \text{Adj Cost}} \tag{44}$$

where A^k, A^n, B^k, B^n are constants, and $\hat{\Sigma}^k$, $\hat{\Sigma}^n$ and $\hat{\Sigma}^{(k,n)}$ follows from Lemma 3.

Proof. See Appendix A.7. □

Same as before, information frictions contribute to the dispersion in MRPK and MRPN via two channels. The first channel stems from the managers' uncertainty about the true productivity, which is summarized by the direct-effect terms (40) and (43). The second channel works through the covariance between the managers' productivity surprises and adjustment-cost wedges, summarized by the interaction term (41) and (44). Using the baseline parameterization, we can quantitatively decompose the MPRK dispersion as

$$\underbrace{0.179}_{var(mrpk)} = \underbrace{1.348}_{\text{Distortion}} + \underbrace{1.321}_{\text{Adj Cost}} + \underbrace{-2.618}_{\text{Adj Cost} \times \text{Distortion}} + \underbrace{0.115}_{\text{Info Friction}} + \underbrace{0.014}_{\text{Info Friction} \times \text{Adj Cost}} \tag{45}$$

and the MPRN dispersion as

$$\underbrace{0.119}_{var(mrpn)} = \underbrace{0.144}_{\text{Distortion}} + \underbrace{0.119}_{\text{Adj Cost}} + \underbrace{-0.250}_{\text{Adj Cost} \times \text{Distortion}} + \underbrace{0.103}_{\text{Info Friction}} + \underbrace{0.002}_{\text{Info Friction} \times \text{Adj Cost}} \tag{46}$$

We can see that information friction is the major source of capital and labor misallocation. The total effect of information friction accounts for 72% of model-implied capital misallocation and about 88% of model-implied labor misallocation. This is due to two reasons. First, the interaction between information friction and adjustment cost is weak and in the same direction as the direct effect. Second, the strong, negative correlation between adjustment wedge and distortion largely offsets the direct contribution of distortions and adjustment costs. Adjustment costs, uncorrelated distortions and information frictions jointly account for about 43% of observed dispersion in capital and about 60% of observed dispersion in labor, resulting in 27.57% of aggregate TFP loss from the efficient benchmark. Information frictions alone can generate about 23% of aggregate TFP loss. This quantification of information-driven misallocation and TFP loss is in line with the estimates in the literature. For instance, David et al. (2016)'s quantification of information-driven TFP loss ranges between 4% and 40%, depending on the internal information structure. My quantification falls in the mid-point of this range.

The decomposition in Proposition 5 does not explicitly show how communication friction contribute to input misallocation and TFP loss. To separate the aggregate effect of communication frictions, I compute the model-implied input misallocation and aggregate TFP loss when communication frictions are removed. Here, I make the capital and labor manager's private signal common knowledge within the firm, so that both managers can use all three signals a_{it}^k, a_{it}^n and a_{it}^p to form expectations. The rest of parameters are fixed at the baseline calibration. We can see that removing the communication friction lowers the dispersion in MRPK by 0.02 and the dispersion in MRPN by 0.008, which improves aggregate TFP by about 2.40%. This modest aggregate effect of communication friction on allocative efficiency is largely due to our previous finding that communication frictions cannot generate large dispersion in capital-labor ratio under the current parameterization.

Table 10: Strategic Complementarity and TFP Gain

	$\epsilon = 4$	$\epsilon = 6$	$\epsilon = 9$
Agg. TFP Loss	-27.57%	-53.00%	-94.67%
Agg. TFP Loss, without communication frictions	-25.17%	-48.59%	-87.37%
TFP gain	2.40%	4.41%	7.31%

The gain of aggregate TFP from removing communication frictions is increasing with the degree of strategic complementarity. Table 10 shows that as the elasiticity of substitution ϵ goes up from 4 (our baseline parameterization) to 9, the aggregate TFP gain goes up from 2.4% to 7.31%. The intuition is similar to what we discussed for the capital-labor ratio: as the capital and labor becomes stronger strategic complement, uncertainty about the other manager's information and decision generates larger dispersion in MRPK and MRPN and hence results in larger TFP losses. Removing communication frictions will largely eliminate the allocative inefficiency from the strategic uncertainty between managers.

4.3 Robustness

Now I explore the robustness of baseline results to alternative mechanisms. In particular, I analyze whether incorporating other frictions in the timing of capital and labor adjustment changes the main messages we established in previous sections: that communication friction is a powerful mechanism of lowering investment-hiring correlation, and that it can generate modest TFP losses.

Time-to-build Our baseline model does not incorporate time-to-build in capital or labor in the estimation. Here, I estimate an extended model with time-to-build in capital and tabulate the results in Table A1 and Table A2 in Appendix A.8.1. Under this setup, the productive capital in today's revenue function is predetermined, and the benefit of today's investment will be realized tomorrow. This creates a timing mismatch for the capital and labor managers when they are planning for capital and labor adjustments.

The first observation is that firm-level common signal plays a more important role in managers' decision-making once we control for time-to-build. The firm-level common signal is estimated to be more precise than the labor manager's private signal and the capital manager's private signal. The reason for this can be seen in Table A3. While information frictions alone drive investment-hiring correlation down to 0.56 (which is close to the data, 0.52), time-to-build is another powerful mechanism to give us investment-hiring disconnect: with time-to-build alone, the investment-hiring correlation is 0.63, and none of other frictions alone can drive this correlation below 0.8. Hence, incorporating time-to-build will reduce the quantitative importance of communication frictions in matching the low investment-hiring correlation.

The second observation is that time-to-build makes the efficiency gain of removing communication frictions mild and smaller than the baseline estimates. Making managers' private signals common knowledge within the firm lowers MRPK dispersion by 0.007 and MPRN dispersion by 0.001, which translates to an aggregate TFP gain of 0.6%. This is because of two reasons. First, the estimated common signal is precise, and the revelation of managers' private signals is not as informative as in our baseline model. Moreover, introducing time-to-build weakens the degree of strategic complementarity between capital and labor. Under time-to-build, today's labor decision is not relevant to the capital manager's investment today because the strategic complementarity between them through the revenue function gone. And as a result, revealing the labor manager's private signal today to the capital manager is not as informative as in the baseline calibration.

Non-convex Cost So far, our calibration is based upon a linear-quadratic setup under convex adjustment cost. Are these results robust to non-linear decision rules and non-convex adjustment costs?

To answer this question, I incorporate non-convex costs in capital and labor adjustment into the model.[10] Following Cooper and Haltiwanger (2006) and Bloom (2009), I assume that the non-convex costs scale with revenue: when the capital manager decides to make an investment, he needs to pay a fixed cost $\chi_k R(A_{it}, K_{it}, N_{it})$, and similarly, the labor manager needs to pay a fixed cost $\chi_n R(A_{it}, K_{it}, N_{it})$ if she decides to adjust labor. To estimate the non-convex cost parameter χ_k and χ_n, I add the inaction rate of capital and labor, which are both about 9 percent in the US data, as two additional targets.[11] In the calibration, I solve the manager's nonlinear decision rule with the true profit function, simulate a panel of 1000 firms for 500 periods, compute the simulated moments and minimize the quadratic loss as before.

The estimated parameters and model fit are tabulated in Table A4 and Table A5 in the Appendix. The non-linear model fits the targeted moments well with small estimated parameters for non-convex costs. Each capital and labor adjustment costs the firm 3 and 2 basis points of revenue respectively, which is significantly smaller than the magnitude reported in the literature. There are two reasons for this small estimate. First, the inaction rate in this Compustat sample is smaller than other samples (such as the Census establishment-level data in Cooper and Haltiwanger (2006)). Second, communication frictions provide a pow-

[10] Under non-convex costs, the MPBE characterization in Lemma 3 is no longer valid because non-convex costs make the decision rules non-Gaussian. To proceed, I assume that the belief updating of capital and labor managers still follow the equations in Lemma 2. Hence, the estimation results should be regarded as an approximated solution to this dynamic team production problem with non-convex costs and incomplete information. See Appendix A.8.4 for details.

[11] In Compustat Fundamentals Annual, very few observations report literally zero investment or hiring. Here, I designate an observation as "inaction" in capital (labor) if the annual absolute percentage change in capital stock (labor) is below 1 percent.

erful mechanism to match the low investment-hiring correlation in the data, which needs to be achieved by non-convex cost if we assume perfect communication. In addition, we can see from Table A4 that the estimated information parameters under non-convex costs are similar to those in the baseline calibration. This indicates that the inferred within-firm communication pattern is robust to the specification of adjustment costs.

Table A4 also shows that non-convex costs do not change the estimates of other parameters significantly. Hence, conclusions from the counterfactual analysis with non-convex costs are largely the same as those from our baseline model. First, communication frictions remain the most prominent force in driving down the investment-hiring correlation. Table A6 shows that non-convex cost alone cannot effectively lower the investment-hiring correlation. This is largely because our small estimates of non-convex costs are not able to reproduce strong asynchronous adjustments in capital and labor. Second, non-convex costs do not affect the quantification of allocative efficiency results. Removing communication frictions lowers MRPK dispersion from 0.177 to 0.154 and MRPN dispersion from 0.099 to 0.093, which jointly generate an aggregate TFP gain of 2.43%, almost identical to the baseline estimates without non-convex costs.

5 Conclusion

This paper studies whether and how within-firm communication friction matters for firm dynamics. It makes three contributions. First, it develops a parsimonious framework to incorporate communication friction into the analysis of firm dynamics. Second, it proposes a novel method to test whether a firm's capital and labor decisions are based on same information or not, using managerial expectation data. I show that the investment forecast errors are simultaneously overreacting to past investment and underreacting to past hiring. I show that this opposite-sign pattern is consistent with the predictions of models with communication frictions and cannot be easily reconciled by models with common information. Lastly, this paper quantifies the aggregate implications of communication frictions for firm dynamics. I estimate a quantitative model of dynamic input choice under communication friction that jointly accounts for the opposite-sign predictability pattern of investment forecast error, high serial correlation of input adjustment, low volatility of input adjustment, and low investment-hiring correlations. The estimated model conveys three important messages. First, manager-level private signals under communication friction is powerful in generating low investment-hiring correlations. Second, although communication friction introduces a new distortion to the capital-labor ratio, its quantitative contribution to the dispersion of capital-labor ratio is small, due to an offsetting force that interacts with adjustment frictions. Lastly, removing communication frictions generates a modest 2.4% of TFP gain, even though information friction is the main source of capital and labor misallocation among the within-firm frictions we consider.

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A Theory Appendix

A.1 Second-order Approximation of the Profit Function

Denote the frictionless steady state value of variable X_t as $\bar{X}$, and denote the log deviation of X_t from $\bar{X}$ as x_t.

Frictionless Steady State Fix $\bar{A} = 1$. The profit maximization problem without information and adjustment frictions is to choose K and N to maximize

$$\Pi(K, N) = Y^{1/\epsilon} A^{1-1/\epsilon} (K^{\alpha} N^{1-\alpha})^{1-1/\epsilon} - RK - WN$$

FOC:

$$Y^{1/\epsilon} A^{1-1/\epsilon} K^{\alpha(1-1/\epsilon)-1} N^{(1-\alpha)(1-1/\epsilon)} \alpha(1 - 1/\epsilon) = R$$

$$Y^{1/\epsilon} A^{1-1/\epsilon} K^{\alpha(1-1/\epsilon)} N^{(1-\alpha)(1-1/\epsilon)-1} (1 - \alpha)(1 - 1/\epsilon) = W$$

In the quantitative exercise, $\alpha, \epsilon, \rho, \sigma_u, \delta, \beta$ are calibrated outside the model. Given these parameters, the frictionless steady-state values for rental rate, capital, labor and wage can be expressed analytically. First, rental price $\bar{R}$ is

$$\bar{R} = 1/\gamma - 1 + \delta = \alpha(1 - 1/\epsilon) \bar{K}^{\alpha(1-1/\epsilon)-1} N^{(1-\alpha)(1-1/\epsilon)}$$

Normalize $\bar{Y} = 1, \bar{A} = 1$, we get

$$\bar{K}^{\alpha(1-1/\epsilon)} \bar{N}^{(1-\alpha)(1-1/\epsilon)} = 1$$

We can use the above two equations, together with FOCs, to back out steady state $\bar{K}, \bar{N}, \bar{W}$:

$$\bar{K} = \frac{\alpha(1 - 1/\epsilon)}{1/\gamma - 1 + \delta}$$

$$\bar{N} = \left(\frac{1/\gamma - 1 + \delta}{\alpha(1 - 1/\epsilon)} \right)^{\alpha/(1-\alpha)}$$

$$\bar{W} = \bar{K}^{\alpha} \bar{N}^{-\alpha} (1 - \alpha)(1 - 1/\epsilon)$$

Second-order Approximation We have

$$
\begin{aligned}
\Pi \approx\ & \bar{A}^{1-1/\epsilon}\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)} + \bar{R}\bar{K}k_t + \bar{W}\bar{N}n_t \\
& + \frac{1}{2}\bar{R}\bar{K}(1-1+\alpha(1-1/\epsilon))k_t^2 + \frac{1}{2}\bar{W}\bar{N}(1-1+(1-\alpha)(1-1/\epsilon))n_t^2 \\
& + \bar{A}^{1-1/\epsilon}\alpha(1-\alpha)(1-1/\epsilon)^2\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}k_t n_t \\
& - \bar{R}\bar{K} - \bar{R}\bar{K}k_t - \bar{R}\bar{K}\frac{1}{2}k_t^2 \\
& - \bar{W}\bar{N} - \bar{W}\bar{N}n_t - \bar{W}\bar{N}\frac{1}{2}n_t^2 \\
& - \frac{\xi_k}{2}\bar{K}(k_t - k_{t-1})^2 - \frac{\xi_n}{2}\bar{N}(n_t - n_{t-1})^2 \\
& + \frac{\epsilon-1}{\epsilon}\bar{A}^{1-1/\epsilon}\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}(a_t + \frac{1}{2}a_t^2) - \frac{1}{2}\frac{\epsilon-1}{\epsilon^2}\bar{A}^{1-1/\epsilon}\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}a_t^2 \\
& + \frac{\epsilon-1}{\epsilon}\bar{A}^{1-1/\epsilon}\alpha(1-1/\epsilon)\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}k_t a_t \\
& + \frac{\epsilon-1}{\epsilon}\bar{A}^{1-1/\epsilon}(1-\alpha)(1-1/\epsilon)\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}n_t a_t
\end{aligned}
$$

where the sixth line is an approximation of adjustment cost for capital:

$$
\begin{aligned}
\frac{\xi_k}{2}\left(\frac{K_t-(1-\delta)K_{t-1}}{K_{t-1}} - \delta\right)^2 K_{t-1} &= \frac{\xi_k}{2}\left(\exp(k_t - k_{t-1}) - 1\right)^2 \bar{K}\exp(k_{t-1}) \\
&= \frac{\xi_k\bar{K}}{2}(k_t - k_{t-1})^2(1 + k_{t-1} + ...) = \frac{\xi_k\bar{K}}{2}(k_t - k_{t-1})^2
\end{aligned}
$$

Define investment $\iota_t \equiv k_t - k_{t-1}$ and hiring $h_t \equiv n_t - n_{t-1}$. Replace $k_t = k_{t-1} + \iota_t$ and $n_t = h_t + n_{t-1}$, we get:

Hence

$$
\begin{aligned}
\pi =\ & \frac{1}{\epsilon}\bar{Y} + \frac{\epsilon-1}{\epsilon}\bar{Y}a_t + \frac{1}{2}\frac{(\epsilon-1)^2}{\epsilon^2}\bar{Y}a_t^2 \\
& + \frac{1}{2}\bar{R}\bar{K}(\alpha(1-1/\epsilon)-1)\left(k_{t-1}^2 + 2k_{t-1}\iota_t + \iota_t^2\right) \\
& + \frac{1}{2}\bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1)(n_{t-1}^2 + 2n_{t-1}h_t + h_t^2) \\
& + \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y}(k_{t-1}n_{t-1} + \iota_t n_{t-1} + k_{t-1}h_t + \iota_t h_t) \\
& + \frac{\epsilon-1}{\epsilon}\alpha(1-1/\epsilon)\bar{Y}(k_{t-1}a_t + \iota_t a_t) + \frac{\epsilon-1}{\epsilon}(1-\alpha)(1-1/\epsilon)\bar{Y}(n_{t-1}a_t + h_t a_t) \\
& - \frac{\xi_k}{2}\bar{K}\iota_t^2 - \frac{\xi_n}{2}\bar{N}h_t^2
\end{aligned}
\tag{A1}
$$

Hence the log-quadratic profit function takes the form

$$x_t'Px_t + x_t'Q\iota_t + x_t'Rh_t + H_\iota\iota_t^2 + H_{\iota h}\iota_t h_t + H_h h_t^2 \tag{A2}$$

where state vector $x_t = [1, k_{t-1}, n_{t-1}, a_t]$, and the matrices are

$$P = \begin{bmatrix} \frac{1}{\epsilon}\bar{Y} & 0 & 0 & \frac{\epsilon-1}{2\epsilon}\bar{Y} \\ * & \frac{1}{2}\bar{R}\bar{K}(\alpha(1-1/\epsilon)-1) & \frac{1}{2}\alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} & \frac{\epsilon-1}{2\epsilon}\alpha(1-1/\epsilon)\bar{Y} \\ * & * & \frac{1}{2}\bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) & \frac{\epsilon-1}{2\epsilon}(1-\alpha)(1-1/\epsilon)\bar{Y} \\ * & * & * & \frac{1}{2}\frac{(\epsilon-1)^2}{\epsilon^2}\bar{Y} \end{bmatrix}$$

$$Q = \begin{bmatrix} 0 \\ \bar{R}\bar{K}(\alpha(1-1/\epsilon)-1) \\ \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} \\ \frac{\epsilon-1}{\epsilon}\alpha(1-1/\epsilon)\bar{Y} \end{bmatrix}$$

$$R = \begin{bmatrix} 0 \\ \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} \\ \bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) \\ \frac{\epsilon-1}{\epsilon}(1-\alpha)(1-1/\epsilon)\bar{Y} \end{bmatrix}$$

and $H_\iota = \frac{1}{2}\bar{R}\bar{K}(\alpha(1-1/\epsilon)-1) - \frac{1}{2}\xi_k\bar{K}$, $H_h = \frac{1}{2}\bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) - \frac{1}{2}\xi_n\bar{N}$, $H_{\iota h} = \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y}$.

The law of motion for the state vector x_t is given by

$$x_{t+1} \equiv \begin{bmatrix} 1 \\ k_{t-1}+\iota_t \\ n_{t-1}+h_t \\ \rho a_t + \mu_{t+1} \end{bmatrix} = Ax_t + B\iota_t + Ch_t + D\mu_{t+1}$$

where

$$A \equiv \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \rho \end{bmatrix}, B \equiv \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, C \equiv \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, D \equiv \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

A.2 Proof of Lemma 2

To save some notations, let

$$\hat{a}_t^k \equiv E\left[a_{it} \middle| \mathcal{I}_{it}^k\right], \hat{\Sigma}_t^k \equiv E\left[(a_{it} - \hat{a}_t^k)(a_{it} - \hat{a}_t^k)'\right]$$

$$\hat{a}_t^n \equiv E\left[a_{it} \middle| \mathcal{I}_{it}^n\right], \hat{\Sigma}_t^n \equiv E\left[(a_{it} - \hat{a}_t^n)(a_{it} - \hat{a}_t^n)'\right]$$

and

$$\hat{a}_t^{(k,n)} \equiv E\left[\hat{a}_t^n \middle| \mathcal{I}_{it}^k\right], \hat{\Sigma}_t^{(k,n)} \equiv E\left[(\hat{a}_t^{(k,n)} - \hat{a}_t^n)(\hat{a}_t^{(k,n)} - \hat{a}_t^n)'\right]$$

$$\hat{a}_t^{(n,k)} \equiv E\left[\hat{a}_t^k \middle| \mathcal{I}_{it}^n\right], \hat{\Sigma}_t^{(n,k)} \equiv E\left[(\hat{a}_t^{(n,k)} - \hat{a}_t^k)(\hat{a}_t^{(n,k)} - \hat{a}_t^k)'\right]$$

so that

$$a_{it} \middle| \mathcal{I}_{it}^k \sim N(\hat{a}_t^k, \hat{\Sigma}_t^k) \text{ and } a_{it} \middle| \mathcal{I}_{it}^n \sim N(\hat{a}_t^n, \hat{\Sigma}_t^n) \tag{A3}$$

characterize the first-order beliefs of capital and labor manager on the state a_{it}, and

$$\hat{a}_t^n \middle| \mathcal{I}_{it}^k \sim N(\hat{a}_t^{(k,n)}, \hat{\Sigma}_t^{(k,n)}) \text{ and } \hat{a}_t^k \middle| \mathcal{I}_{it}^n \sim N(\hat{a}_t^{(n,k)}, \hat{\Sigma}_t^{(n,k)}) \tag{A4}$$

characterize the high-order beliefs of capital and labor manager on each other's beliefs about a_{it}.

Lemma 2 states that the first-order and higher-order uncertainty are given by the following triplet of Riccati equations:

$$\begin{aligned}
\hat{\Sigma}_t^k &= (\hat{\Sigma}_t^k)^- - (\hat{\Sigma}_t^k)^- H' \left[H(\hat{\Sigma}_t^k)^- H' + W^k\right]^{-1} H(\hat{\Sigma}_t^k)^- && \text{(A5)} \\
\hat{\Sigma}_t^n &= (\hat{\Sigma}_t^n)^- - (\hat{\Sigma}_t^n)^- H' \left[H(\hat{\Sigma}_t^n)^- H' + W^n\right]^{-1} H(\hat{\Sigma}_t^n)^- && \text{(A6)} \\
\hat{\Sigma}_t^{(k,n)} &= \hat{\Sigma}_t^k + \hat{\Sigma}_t^n - 2\tilde{\Sigma}_t^{(k,n)} && \text{(A7)}
\end{aligned}$$

where the pre-estimate uncertainty $(\hat{\Sigma}_t^k)^-, (\hat{\Sigma}_t^n)^-$ are given by

$$(\hat{\Sigma}_{t-1}^k)^+ = \hat{\Sigma}_{t-1}^k - (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)}) \left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1} (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})' \tag{A8}$$

$$\begin{aligned}
(\hat{\Sigma}_t^k)^- &= \rho^2 \left\{ \hat{\Sigma}_{t-1}^k - (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)}) \left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1} (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})' \right\} + \sigma_\mu^2 \\
(\hat{\Sigma}_t^n)^- &= \rho^2 \left\{ \hat{\Sigma}_{t-1}^n - (\hat{\Sigma}_{t-1}^n - \tilde{\Sigma}_{t-1}^{(k,n)}) \left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1} (\hat{\Sigma}_{t-1}^n - \tilde{\Sigma}_{t-1}^{(k,n)})' \right\} + \sigma_\mu^2
\end{aligned}$$

and the covariance between forecast errors $\tilde{\Sigma}_t^{(k,n)}$ is defined recursively by

$$\begin{aligned}
\tilde{\Sigma}_t^{(k,n)} = &\left(I - (\hat{\Sigma}_t^k)^- H' \left[H(\hat{\Sigma}_t^k)^- H' + W^k \right]^{-1} H \right) \Lambda_t \left(I - (\hat{\Sigma}_t^n)^- H' \left[H(\hat{\Sigma}_t^n)^- H' + W^n \right]^{-1} H \right)' \\
&+ (\hat{\Sigma}_t^k)^- H' \left[H(\hat{\Sigma}_t^k)^- H' + W^k \right]^{-1} \Omega_t \left[H(\hat{\Sigma}_t^n)^- H' + W^n \right]^{-1} H(\hat{\Sigma}_t^n)^-
\end{aligned} \tag{A9}$$

with

$$\Lambda_t = \rho^2 \left\{ \tilde{\Sigma}_{t-1}^{(k,n)} + (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)}) \left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1} (\hat{\Sigma}_{t-1}^n - \tilde{\Sigma}_{t-1}^{(k,n)}) \right\} + \sigma_\mu^2$$

Initial Beliefs To fully specify the belief dynamics, I assume that the capital manager believes that the initial state $a_{i,0}$ is drawn from a Gaussian distribution

$$a_{i,0} \sim N(\hat{a}_0^k, W_0^k)$$

and that the initial information set $\mathcal{I}_0^k = \{\hat{a}_0^k\}$. Similarly, for the labor manager, she believes that the initial state is drawn from a Gaussian distribution

$$a_{i,0} \sim N(\hat{a}_0^n, W_0^n)$$

and her initial information set is $\mathcal{I}_0^n = \{\hat{a}_0^n\}$. For simplicity, I assume that the two initial distributions are independent to each other.

For simplicity, I assume that W_0^k and W_0^n are common knowledge. To characterize the initial higher-order belief $\hat{a}_0^n \big| \mathcal{I}_0^k$ and $\hat{a}_0^k \big| \mathcal{I}_0^n$, it is perhaps more convenient to define the initial forecast errors e_0^k, e_0^n as

$$e_0^k \equiv \hat{a}_0^k - a_{i,0} \text{ and } e_0^n \equiv \hat{a}_0^n - a_{i,0}$$

with $e_0^k \sim N(0, W_0^k)$ and $e_0^n \sim N(0, W_0^n)$ being independent to each other. Therefore

$$\hat{a}_0^n = \hat{a}_0^k - e_0^k + e_0^n$$

From the capital manager's perspective,

$$E\left[\hat{a}_0^n \middle| \mathcal{I}_0^k\right] = \hat{a}_0^k$$

$$var\left[\hat{a}_0^n \middle| \mathcal{I}_0^k\right] = var(-e_0^k + e_0^n) = W_0^n + W_0^k$$

Therefore the initial high-order beliefs are

$$\hat{a}_0^n \middle| \mathcal{I}_0^k \sim N(\hat{a}_0^k, W_0^n + W_0^k) \text{ and } \hat{a}_0^k \middle| \mathcal{I}_0^n \sim N(\hat{a}_0^n, W_0^n + W_0^k)$$

A first-glance at HOB at period t The same method can be used to characterize the first moment of high-order beliefs at period t. At period t, from the capital manager's perspective,

$$\hat{a}_t^n = \hat{a}_t^k - e_t^k + e_t^n$$

where e_t^k and e_t^n are forecast errors of the capital and labor manager. Hence

$$\hat{a}_t^{(k,n)} = E\left[\hat{a}_t^n \middle| \mathcal{I}_t^k\right] = \hat{a}_t^k \tag{A10}$$

$$\hat{a}_t^{(n,k)} = E\left[\hat{a}_t^k \middle| \mathcal{I}_t^n\right] = \hat{a}_t^n \tag{A11}$$

Characterizing the second moment of high-order belief is a bit more involved since the forecast errors e_t^k and e_t^n are correlated with each other. For simplicity, I denote the correlation $E[e_t^k e_t^n]$ as $\tilde{\Sigma}_t^{(k,n)}$. Immediately, we have

$$\hat{\Sigma}_t^{(k,n)} = E\left[(\hat{a}_t^n - \hat{a}_t^{(k,n)})^2\right] = var\left[-e_t^k + e_t^n\right] = \hat{\Sigma}_t^k + \hat{\Sigma}_t^n - 2\tilde{\Sigma}_t^{(k,n)} \tag{A12}$$

Partitioning the State Vector Note that our state vector $x_t \equiv [1, k_{t-1}, n_{t-1}, a_{it}]'$ can be partitioned into a perfectly observed part $s_t \equiv [1, k_{i,t-1}, n_{i,t-1}]'$ and an imperfectly observed part a_{it}. Hence, the linear policy

$$\iota_t = F_\iota E\left[x_t \middle| \mathcal{I}_t^k\right], h_t = F_h E\left[x_t \middle| \mathcal{I}_t^n\right]$$

can be rewritten as

$$\iota_t = F_\iota^s s_t + F_\iota^a \hat{a}_t^k \tag{A13}$$
$$h_t = F_h^s s_t + F_h^a \hat{a}_t^n \tag{A14}$$

Now I specify the observation equations for the partitioned state vector. The capital manager's signals about a_{it} are summarized by

$$z_t^k \equiv \begin{bmatrix} a_t^p \\ a_t^k \end{bmatrix} = Ha_t + w_t^k \tag{A15}$$

where $H = [1,1]'$ and $w_t^k = [\epsilon_t^p, \epsilon_t^k]' \sim N(0, diag(\sigma_p^2, \sigma_{\epsilon,k}^2))$. Similarly, the labor manager's signals about a_{it} are summarized by

$$z_t^n \equiv \begin{bmatrix} a_t^p \\ a_t^n \end{bmatrix} = Ha_t + w_t^n \tag{A16}$$

where $H = [1,1]'$ and $w_t^k = [\epsilon_t^p, \epsilon_t^n]' \sim N(0, diag(\sigma_p^2, \sigma_{\epsilon,n}^2))$.

Updating FOB and HOBs Now, suppose we are at the end of period $t-1$, with information set $\mathcal{I}_{t-1}^k$ and $\mathcal{I}_{t-1}^n$. From now on I will only focus on how the capital manager updates beliefs, given he knows that the labor manager's action takes the form

$$h_t = F_h^s s_t + F_h^a \hat{a}_t^n$$

the arguments for the labor manager are basically the same.

At period t, the belief updating takes two steps:

1. The capital manager observes the labor manager's action $h_{t-1} = F_h^s s_{t-1} + F_h^a \hat{a}_{t-1}^n$ and updates the estimates about a_{t-1} as well as a_t. Formally, after observing h_{t-1}, the capital manager computes

$$a_{t-1} \big| \mathcal{I}_{t-1}^k \cup \{h_{t-1}\} \sim N\left((\hat{a}_{t-1}^k)^+, (\hat{\Sigma}_{t-1}^k)^+\right)$$

and

$$a_t \big| \mathcal{I}_{t-1}^k \cup \{h_{t-1}\} \sim N\left((\hat{a}_t^k)^-, (\hat{\Sigma}_t^k)^-\right)$$

2. Then the capital manager uses the private signal z_t^k to further update the state estimate a_t on top of the pre-estimate $a_t\big|\mathcal{I}_{t-1}^k \cup \{h_{t-1}\}$ and get

$$a_t\big|\mathcal{I}_{t-1}^k \cup \{h_{t-1}\} \cup \{z_t^k\} = a_t\big|\mathcal{I}_t^k \sim N\left(\hat{a}_t^k, \hat{\Sigma}_t^k\right)$$

We do it step-by-step.

The first step: note that

$$\begin{bmatrix} a_{t-1} \\ h_{t-1} \end{bmatrix} \bigg|\mathcal{I}_{t-1}^k = \begin{bmatrix} a_{t-1} \\ F_h^s s_{t-1} + F_h^a \hat{a}_{t-1}^n \end{bmatrix} \bigg|\mathcal{I}_{t-1}^k \sim N\left(\begin{bmatrix} \hat{a}_{t-1}^k \\ F_h^s s_{t-1} + F_h^a \hat{a}_{t-1}^{(k,n)} \end{bmatrix}, \begin{bmatrix} \hat{\Sigma}_{t-1}^k & * \\ * & F_h^a \hat{\Sigma}_{t-1}^{(k,n)} (F_h^a)' \end{bmatrix}\right) \tag{A17}$$

where $*$ is equal to

$$\begin{aligned} * &= E\left[(a_{t-1} - \hat{a}_{t-1}^k)(\hat{a}_{t-1}^n - \hat{a}_{t-1}^{(k,n)})'(F_h^a)'\right] = E\left[(a_{t-1} - \hat{a}_{t-1}^k)(\hat{a}_{t-1}^n - \hat{a}_{t-1}^k)'(F_h^a)'\right] \\ &= E\left[-e_{t-1}^k(\hat{a}_{t-1}^n - a_{t-1} + a_{t-1} - \hat{a}_{t-1}^k)'(F_h^a)'\right] = E\left[-e_{t-1}^k(e_{t-1}^n - e_{t-1}^k)'(F_h^a)'\right] \\ &= (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})(F_h^a)' \end{aligned} \tag{A18}$$

Therefore the capital manager updates his belief about a_{t-1} using the labor manager's action by

$$(\hat{a}_{t-1}^k)^+ = \hat{a}_{t-1}^k + (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})(F_h^a)'\left[F_h^a \hat{\Sigma}_{t-1}^{(k,n)} (F_h^a)'\right]^{-1} F_h^a(\hat{a}_{t-1}^n - \hat{a}_{t-1}^k) \tag{A19}$$

$$(\hat{\Sigma}_{t-1}^k)^+ = \hat{\Sigma}_{t-1}^k - (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})(F_h^a)'\left[F_h^a \hat{\Sigma}_{t-1}^{(k,n)} (F_h^a)'\right]^{-1} F_h^a(\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})' \tag{A20}$$

Since F_h^a is a scalar, these can be simplified as

$$(\hat{a}_{t-1}^k)^+ = \hat{a}_{t-1}^k + (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}(\hat{a}_{t-1}^n - \hat{a}_{t-1}^k) \tag{A21}$$

$$(\hat{\Sigma}_{t-1}^k)^+ = \hat{\Sigma}_{t-1}^k - (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}(\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})' \tag{A22}$$

and therefore, the pre-estimates for a_t after observing the opponent's action is given by

$$(\hat{a}_t^k)^- = \rho(\hat{a}_{t-1}^k)^+ \tag{A23}$$

$$(\hat{\Sigma}_t^k)^- = \rho^2(\hat{\Sigma}_{t-1}^k)^+ + \sigma_\mu^2 \tag{A24}$$

Now we incorporate the signal z_t^k into the information set and update the belief about a_t.

We have

$$\begin{bmatrix} a_t \\ z_t^k \end{bmatrix} \Big| \mathcal{I}_{t-1}^k \cup \{h_{t-1}\} \sim N\left(\begin{bmatrix} (\hat{a}_t^k)^- \\ H(\hat{a}_t^k)^- \end{bmatrix}, \begin{bmatrix} (\hat{\Sigma}_t^k)^- & (\hat{\Sigma}_t^k)^- H' \\ H(\hat{\Sigma}_t^k)^- & H(\hat{\Sigma}_t^k)^- H' + W^k \end{bmatrix} \right)$$

where $W^k \equiv diag(\sigma_p^2, \sigma_{\epsilon,k}^2)$. Therefore,

$$\hat{a}_t^k = (\hat{a}_t^k)^- + (\hat{\Sigma}_t^k)^- H' \left[H(\hat{\Sigma}_t^k)^- H' + W^k \right]^{-1} \left(z_t^k - H(\hat{a}_t^k)^- \right) \tag{A25}$$

$$\hat{\Sigma}_t^k = (\hat{\Sigma}_t^k)^- - (\hat{\Sigma}_t^k)^- H' \left[H(\hat{\Sigma}_t^k)^- H' + W^k \right]^{-1} H(\hat{\Sigma}_t^k)^- \tag{A26}$$

To close the analysis, I need to specify how high-order beliefs evolve. According to (A12), the uncertainty of the capital manager on the labor manager's expectation about the productivity is a function of (1) the capital manager's posterior uncertainty $\hat{\Sigma}_t^k$ on the productivity at period t, (2) the labor manager's posterior uncertainty $\hat{\Sigma}_t^n$ on the productivity at period t, and (3) the covariance between capital manager and labor manager's forecast errors $\tilde{\Sigma}_t^{(k,n)}$. We already have the law of motion for $\hat{\Sigma}_t^k$ given by (A26) and a similar law of motion for $\hat{\Sigma}_t^n$. It remains to characterize the covariance for forecast errors $\tilde{\Sigma}_t^{(k,n)}$.

According to (A25), the forecast error at period t is given by

$$\hat{a}_t^k - a_t = \left(I - (\hat{\Sigma}_t^k)^- H' \left[H(\hat{\Sigma}_t^k)^- H' + W^k \right]^{-1} H \right) \left((\hat{a}_t^k)^- - a_t \right) + (\hat{\Sigma}_t^k)^- H' \left[H(\hat{\Sigma}_t^k)^- H' + W^k \right]^{-1} w_t^k \tag{A27}$$

and similarly

$$\hat{a}_t^n - a_t = \left(I - (\hat{\Sigma}_t^n)^- H' \left[H(\hat{\Sigma}_t^n)^- H' + W^n \right]^{-1} H \right) \left((\hat{a}_t^n)^- - a_t \right) + (\hat{\Sigma}_t^n)^- H' \left[H(\hat{\Sigma}_t^n)^- H' + W^n \right]^{-1} w_t^n \tag{A28}$$

Therefore

$$\begin{aligned} \tilde{\Sigma}_t^{(k,n)} = & \left(I - (\hat{\Sigma}_t^k)^- H' \left[H(\hat{\Sigma}_t^k)^- H' + W^k \right]^{-1} H \right) \Lambda_t \left(I - (\hat{\Sigma}_t^n)^- H' \left[H(\hat{\Sigma}_t^n)^- H' + W^n \right]^{-1} H \right)' \\ & + (\hat{\Sigma}_t^k)^- H' \left[H(\hat{\Sigma}_t^k)^- H' + W^k \right]^{-1} \Omega_t \left[H(\hat{\Sigma}_t^n)^- H' + W^n \right]^{-1} H(\hat{\Sigma}_t^n)^- \end{aligned} \tag{A29}$$

where $\Lambda_t \equiv E\left[(a_t - (\hat{a}_t^k)^-)(a_t - (\hat{a}_t^n)^-)' \right]$ and $\Omega_t = E\left[w_t^k (w_t^n)' \right] = diag(\sigma_p^2, 0)$.

We know that

$$a_t - (\hat{a}_t^k)^- = \rho a_{t-1} + \mu_t - \rho(\hat{a}_{t-1}^k)^+ = \rho(a_{t-1} - (\hat{a}_{t-1}^k)^+) + \mu_t$$

$$a_t - (\hat{a}_t^n)^- = \rho a_{t-1} + \mu_t - \rho(\hat{a}_{t-1}^n)^+ = \rho(a_{t-1} - (\hat{a}_{t-1}^n)^+) + \mu_t$$

and therefore

$$\Lambda_t = E\left[(a_t - (\hat{a}_t^k)^-)(a_t - (\hat{a}_t^n)^-)'\right] = \rho^2 E\left[(a_{t-1} - (\hat{a}_{t-1}^k)^+)(a_{t-1} - (\hat{a}_{t-1}^n)^+)'\right] + \sigma_\mu^2$$

It remains to compute $E\left[(a_{t-1} - (\hat{a}_{t-1}^k)^+)(a_{t-1} - (\hat{a}_{t-1}^n)^+)'\right]$. Note that

$$a_{t-1} - (\hat{a}_{t-1}^k)^+ = -\left(I - (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}\right) e_{t-1}^k - (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1} e_{t-1}^n$$

$$a_{t-1} - (\hat{a}_{t-1}^n)^+ = -\left(I - (\hat{\Sigma}_{t-1}^n - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}\right) e_{t-1}^n - (\hat{\Sigma}_{t-1}^n - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1} e_{t-1}^k$$

Hence

$$\begin{aligned}
&E\left[(a_{t-1} - (\hat{a}_{t-1}^k)^+)(a_{t-1} - (\hat{a}_{t-1}^n)^+)'\right] \\
&= \left(I - (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}\right)\hat{\Sigma}_{t-1}^{(k,n)}\left(I - (\hat{\Sigma}_{t-1}^n - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}\right)' \\
&+ (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}\hat{\Sigma}_{t-1}^n\left(I - (\hat{\Sigma}_{t-1}^n - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}\right)' \\
&+ \left(I - (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}\right)\hat{\Sigma}_{t-1}^k\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}(\hat{\Sigma}_{t-1}^n - \tilde{\Sigma}_{t-1}^{(k,n)})' \\
&+ (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}\hat{\Sigma}_{t-1}^{(k,n)}\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}(\hat{\Sigma}_{t-1}^n - \tilde{\Sigma}_{t-1}^{(k,n)})' \\
&= \tilde{\Sigma}_{t-1}^{(k,n)} + (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}(\hat{\Sigma}_{t-1}^n - \tilde{\Sigma}_{t-1}^{(k,n)}) \qquad \text{(A30)}
\end{aligned}$$

and therefore

$$\Lambda_t = \rho^2\left\{\tilde{\Sigma}_{t-1}^{(k,n)} + (\hat{\Sigma}_{t-1}^k - \tilde{\Sigma}_{t-1}^{(k,n)})\left[\hat{\Sigma}_{t-1}^{(k,n)}\right]^{-1}(\hat{\Sigma}_{t-1}^n - \tilde{\Sigma}_{t-1}^{(k,n)})\right\} + \sigma_\mu^2 \qquad \text{(A31)}$$

and we finally complete our characterization of belief updates.

A.3 Proof of Lemma 1

A.3.1 Part I: linearity of MPE

Guess that ι and h are linear:

$$\iota_t = F_\iota x_t, h_t = F_h x_t$$

Hence, the MPE is defined as

- Given F_h, the capital manager's strategy $\iota_t = F_\iota x_t$ solves the Bellman equation

$$\begin{aligned} V^k(x_t) &= \max_{\iota_t} x_t' P x_t + x_t' Q \iota_t + x_t' R F_h x_t + H_\iota \iota_t^2 + H_{\iota h} \iota_t F_h x_t + x_t' F_h' H_h F_h x_t + \gamma E\left[V^k(x_{t+1})\right] \\ &= \max_{\iota_t} x_t' \left(P + R F_h + F_h' H_h F_h\right) x_t + x_t' \left(Q + F_h' H_{\iota h}\right) \iota_t + H_\iota \iota_t^2 + \gamma E\left[V^k(x_{t+1})\right] \end{aligned} \tag{A32}$$

  subject to the law of motion (15) with h_t replaced by $F_h x_t$:

$$x_{t+1} = \left(A + C F_h\right) x_t + B \iota_t + D \mu_{i,t+1} \tag{A33}$$

- Given F_ι, the labor manager's strategy $h_t = F_h x_t$ solves the Bellman equation

$$\begin{aligned} V^n(x_t) &= \max_{h_t} x_t' P x_t + x_t' Q F_\iota x_t + x_t' R h_t + x_t' F_\iota' H_\iota F_\iota x_t + x_t' F_\iota' H_{\iota h} h_t + H_h h_t^2 + \gamma E\left[V^n(x_{t+1})\right] \\ &= \max_{h_t} x_t' \left(P + Q F_\iota + F_\iota' H_\iota F_\iota\right) x_t + x_t' \left(R + F_\iota' H_{\iota h}\right) h_t + H_h h_t^2 + \gamma E\left[V^n(x_{t+1})\right] \end{aligned} \tag{A34}$$

  subject to the law of motion (15), with ι_t replaced by $F_\iota x_t$:

$$x_{t+1} = \left(A + B F_\iota\right) x_t + C h_t + D \mu_{i,t+1} \tag{A35}$$

With the linear-quadratic setting, we know that the value functions V^k and V^n takes the form

$$V^k(x_t) = x_t' \hat{P}_k x_t + \hat{Q}_k, V^n(x_t) = x_t' \hat{P}_n x_t + \hat{Q}_n$$

Plug it into the Bellman equations: capital manager's problem becomes

$$\begin{aligned} & x_t' \left[P + R F_h + F_h' H_h F_h + \gamma (A + C F_h)' \hat{P}_k (A + C F_h)\right] x_t + i_t' (H_i + \gamma B' \hat{P}_k B) i_t \\ & + i_t' (H_{ih} F_h + Q' + \gamma B' \hat{P}_k (A + C F_h)) x_t + \gamma x_t' (A + C F_h)' \hat{P}_k B i_t \end{aligned}$$

$$= \cdots + i_t' \left[H_{ih} F_h + Q' + \gamma B'(\hat{P}_k + \hat{P}_k')(A + CF_h) \right] x_t$$

take first order condition on ι:

$$2(H_i + \gamma B' \hat{P}_k B)\iota_t = - \left[H_{ih} F_h + Q' + \gamma B'(\hat{P}_k + \hat{P}_k')(A + CF_h) \right] x_t$$

we get the policy matrix F_ι that solves (A32):

$$F_\iota = -\frac{1}{2} \left(H_\iota + \gamma B' \hat{P}_k B \right)^{-1} \left[Q' + H_{\iota h} F_h + \gamma B'(\hat{P}_k + \hat{P}_k')(A + CF_h) \right] \tag{A36}$$

where $\hat{P}_k$ is defined by Riccati equation

$$\begin{aligned} \hat{P}_k &= P + RF_h + F_h' H_h F_h + \gamma (A + CF_h)' \hat{P}_k (A + CF_h) \\ &- \frac{1}{4} \left[Q' + H_{\iota h} F_h + 2\gamma B' \hat{P}_k (A + CF_h) \right]' \left(H_\iota + \gamma B' \hat{P}_k B \right)^{-1} \left[Q' + H_{\iota h} F_h + 2\gamma B' \hat{P}_k (A + CF_h) \right] \end{aligned} \tag{A37}$$

and similarly the solution to (A34) is

$$F_h = -\frac{1}{2} \left(H_h + \gamma C' \hat{P}_n C \right)^{-1} \left[R' + H_{\iota h} F_\iota + 2\gamma C' \hat{P}_n (A + BF_\iota) \right] \tag{A38}$$

where $\hat{P}_n$ is defined by Riccati equation

$$\begin{aligned} \hat{P}_n &= P + QF_\iota + F_\iota' H_\iota F_\iota + (A + BF_\iota)' \hat{P}_n (A + BF_\iota) \\ &- \frac{1}{4} \left[R' + H_{\iota h} F_\iota + 2\gamma C' \hat{P}_n (A + BF_\iota) \right]' \left(H_h + \gamma C' \hat{P}_n C \right)^{-1} \left[R' + H_{\iota h} F_\iota + 2\gamma C' \hat{P}_n (A + BF_\iota) \right] \end{aligned} \tag{A39}$$

A.3.2 Part II: linearity of MPBE

Now that we have characterized beliefs given policy F_ι, F_h, I go back to the Bellman equations and derive these policy matrices F_ι, F_h. Under incomplete information, the capital manager's Bellman equation is

$$V^k(x_t) = \max_{\iota_t} E \left[x_t' \left(P + RF_h + F_h' H_h F_h \right) x_t + x_t' \left(Q + F_h' H_{\iota h} \right) \iota_t + H_\iota \iota_t^2 + \gamma V^k(x_{t+1}) \middle| \mathcal{I}_t^k \right] \tag{A40}$$

To save some notation, let $\tilde{P}^k \equiv P + RF_h + F_h' H_h F_h$ and $\tilde{Q}^k \equiv Q + F_h' H_{\iota h}$.

From the previous section, we know that $x_t = [s_t, a_t]'$ where $s_t \equiv [1, k_{i,t-1}, n_{i,t-1}]'$ is the perfectly observable part, and a_t is the partially observed productivity. We know that the time-t estimate of x_t for the capital manager is

$$\hat{x}_t^k \equiv [s_t, \hat{a}_t^k]'$$

To see how the noisy information changes the Bellman equation, we first look at the first term $x_t' \left(P + RF_h + F_h' H_h F_h\right) x_t$. Put it in the conditional expectation, we get

$$\begin{aligned}
E\left[x_t' \left(P + RF_h + F_h' H_h F_h\right) x_t \middle| \mathcal{I}_t^k\right] &= E\left[x_t' \tilde{P}^k x_t \middle| \mathcal{I}_t^k\right] \\
&= E\left[[s_t', a_t'] \begin{bmatrix} \tilde{P}_{11}^k & \tilde{P}_{12}^k \\ \tilde{P}_{21}^k & \tilde{P}_{22}^k \end{bmatrix} \begin{bmatrix} s_t \\ a_t \end{bmatrix} \middle| \mathcal{I}_t^k\right] \\
&= E\left[s_t' \tilde{P}_{11}^k s_t + s_t' \tilde{P}_{12}^k a_t + a_t' \tilde{P}_{21}^k s_t + a_t' \tilde{P}_{22}^k a_t \middle| \mathcal{I}_t^k\right] \\
&= s_t' \tilde{P}_{11}^k s_t + s_t' \tilde{P}_{12}^k \hat{a}_t^k + (\hat{a}_t^k)' \tilde{P}_{21}^k s_t + E\left[a_t' \tilde{P}_{22}^k a_t \middle| \mathcal{I}_t^k\right]
\end{aligned}$$

We know that

$$\begin{aligned}
E\left[a_t' \tilde{P}_{22}^k a_t \middle| \mathcal{I}_t^k\right] &= E\left[tr\left(a_t' \tilde{P}_{22}^k a_t\right) \middle| \mathcal{I}_t^k\right] \\
&= E\left[tr\left(\tilde{P}_{22}^k a_t a_t'\right) \middle| \mathcal{I}_t^k\right] = tr\left(\tilde{P}_{22}^k E\left[a_t a_t' \middle| \mathcal{I}_t^k\right]\right) \\
&= tr\left(\tilde{P}_{22}^k \left[\hat{a}_t^k (\hat{a}_t^k)' + \hat{\Sigma}_t^k\right]\right) = tr\left(\tilde{P}_{22}^k \hat{\Sigma}_t^k\right) + tr\left(\tilde{P}_{22}^k \hat{a}_t^k (\hat{a}_t^k)'\right) \\
&= tr\left(\tilde{P}_{22}^k \hat{\Sigma}_t^k\right) + (\hat{a}_t^k)' \tilde{P}_{22}^k \hat{a}_t^k
\end{aligned}$$

Plug it back:

$$\begin{aligned}
E\left[x_t' \left(P + RF_h + F_h' H_h F_h\right) x_t \middle| \mathcal{I}_t^k\right] &= s_t' \tilde{P}_{11}^k s_t + s_t' \tilde{P}_{12}^k \hat{a}_t^k + (\hat{a}_t^k)' \tilde{P}_{21}^k s_t + E\left[a_t' \tilde{P}_{22}^k a_t \middle| \mathcal{I}_t^k\right] \\
&= s_t' \tilde{P}_{11}^k s_t + s_t' \tilde{P}_{12}^k \hat{a}_t^k + (\hat{a}_t^k)' \tilde{P}_{21}^k s_t + tr\left(\tilde{P}_{22}^k \hat{\Sigma}_t^k\right) + (\hat{a}_t^k)' \tilde{P}_{22}^k \hat{a}_t^k \\
&= tr\left(\tilde{P}_{22}^k \hat{\Sigma}_t^k\right) + [s_t', (\hat{a}_t^k)'] \begin{bmatrix} \tilde{P}_{11}^k & \tilde{P}_{12}^k \\ \tilde{P}_{21}^k & \tilde{P}_{22}^k \end{bmatrix} \begin{bmatrix} s_t \\ \hat{a}_t^k \end{bmatrix} \\
&= tr\left(\tilde{P}_{22}^k \hat{\Sigma}_t^k\right) + (\hat{x}_t^k)' \tilde{P}^k \hat{x}_t^k
\end{aligned}$$

The rest of terms in the quadratic profit function is straight forward. For the second term,

$$E\left[x_t'\left(Q+F_h'H_{\iota h}\right)\iota_t\middle|\mathcal{I}_t^k\right]=(\hat{x}_t^k)'\tilde{Q}^k\iota_t$$

and the third term can be taken out of the expectation directly since ι_t is a function of elements in $\mathcal{I}_t^k$.

As a result, we have

$$\begin{aligned}V^k(x_t)=\max_{\iota_t} tr\left(\tilde{P}_{22}^k\hat{\Sigma}_t^k\right)+(\hat{x}_t^k)'\left(P+RF_h+F_h'H_hF_h\right)\hat{x}_t^k+(\hat{x}_t^k)'\left(Q+F_h'H_{\iota h}\right)\iota_t+H_\iota\iota_t^2\\+\gamma E\left[V^k(x_{t+1})\middle|\mathcal{I}_t^k\right]\end{aligned}\tag{A41}$$

Compared with the complete-information case, we can see that the expected profit under incomplete information is different in the following aspects:

- replacing the true state variable x_t by the belief $\hat{x}_t^k$;
- adding a constant term that contains the posterior uncertainty $\hat{\Sigma}_t^k$.

Notably, the second moment of state estimate $\hat{\Sigma}_t^k$ enters the value function only via the constant term (in the trace) and does not become part of the coefficient on terms with $\hat{x}_t^k$ or ι_t. Hence, $\hat{\Sigma}_t^k$ will not affect the first-order conditions of the Bellman equation — this should not be surprising since we are doing linear-quadratic control and certainty equivalence holds.

Again, I guess that the value function takes the form

$$(\hat{x}_t^k)'\hat{P}_k\hat{x}_t^k+\hat{Q}_k$$

so that

$$\begin{aligned}E\left[V^k(x_{t+1})\middle|\mathcal{I}_t^k\right]&=E\left[(\hat{x}_{t+1}^k)'\hat{P}_k\hat{x}_{t+1}^k+\hat{Q}_k\middle|\mathcal{I}_t^k\right]\\&=tr\left(\hat{P}_k\hat{\Sigma}_{t+1}^k\right)+(\hat{x}_{t+1}^k)'\hat{P}_k\hat{x}_{t+1}^k+\hat{Q}_k\end{aligned}$$

We plug it into the (A41) and take first order condition, we will end up with exactly the same F_ι that we had in (A36), the full-information benchmark. The undertermined coefficient matrix $\hat{P}_k$ in the value function is pinned down by exactly the same Riccatti equation (A37) in the full-information benchmark. The constant matrix $\hat{Q}^k$ will be different, since now it has to contain the trace term with posterior uncertainty.

Similarly, for the labor manager, the policy matrix F_h will be exactly the same as the (A38), with Riccatti equation defined by (A39).

A.4 Computation of Joint Distribution and Estimation

The computation of SS Joint Distribution of $[k_{it}, n_{it}, \hat{a}^k_{it}, \hat{a}^n_{it}, a_{it}]' \sim N(0, \mathbf{\Sigma})$ takes the following steps:

1. Compute the decision rule F_ι, F_h for capital and labor under the complete-info Linear MPE. Denote $F_\iota = [F^k_\iota, F^n_\iota, F^a_\iota]$ and $F_h = [F^k_h, F^n_h, F^a_h]$.
2. Given F_ι, F_h, use (A5) - (A7) to get the steady-state beliefs $\hat{\Sigma}_k = \lim \hat{\Sigma}^k_t, \hat{\Sigma}_n = \lim \hat{\Sigma}^n_t$ and $\hat{\Sigma}_{(k,n)} = \lim \hat{\Sigma}^{(k,n)}_t$.
3. Stack the decision rule and the belief dynamics into the following recursion:

$$B \begin{bmatrix} \iota_t \\ h_t \\ \hat{a}^k_t \\ \hat{a}^n_t \\ a_t \end{bmatrix} = C \begin{bmatrix} k_{t-1} \\ n_{t-1} \\ \hat{a}^k_{t-1} \\ \hat{a}^n_{t-1} \\ a_{t-1} \end{bmatrix} + D \begin{bmatrix} \epsilon^p_t \\ \epsilon^k_t \\ \epsilon^n_t \\ \mu_t \end{bmatrix}$$

   where elements in B, C and D can be computed from $F_\iota, F_h, \hat{\Sigma}_k, \hat{\Sigma}_n, \hat{\Sigma}_{(k,n)}$.
4. Rewrite the above equation in terms of $[k_{t-1}, n_{t-1}, \hat{a}^k_{t-1}, \hat{a}^n_{t-1}, a_{t-1}]'$:

$$\begin{bmatrix} k_t \\ n_t \\ \hat{a}^k_t \\ \hat{a}^n_t \\ a_t \end{bmatrix} = \tilde{C} \begin{bmatrix} k_{t-1} \\ n_{t-1} \\ \hat{a}^k_{t-1} \\ \hat{a}^n_{t-1} \\ a_{t-1} \end{bmatrix} + \tilde{D} \begin{bmatrix} \epsilon^p_t \\ \epsilon^k_t \\ \epsilon^n_t \\ \mu_t \end{bmatrix}$$

   where $\tilde{C} \equiv B^{-1}C + diag(1, 1, 0, 0, 0), \tilde{D} \equiv B^{-1}D$.
5. Compute the variance-covariance matrix of $[k_{t-1}, n_{t-1}, \hat{a}^k_{t-1}, \hat{a}^n_{t-1}, a_{t-1}]'$, denoted as $\mathbf{\Sigma}$, by Lyapunov equation

$$\mathbf{\Sigma} = \tilde{C}\mathbf{\Sigma}\tilde{C}' + \tilde{D}\Sigma_\epsilon\tilde{D}'$$

solve this Lyapunov equation and we obtain $\mathbf{\Sigma}$.

A.5 Proof of Proposition 1

We know that

$$\begin{aligned} \iota_t &= F_\iota^k k_{t-1} + F_\iota^n n_{t-1} + F_\iota^a \hat{a}_t^k \\ \mathbb{E}_{t-1}^k[\iota_t] &= F_\iota^k k_{t-1} + F_\iota^n \mathbb{E}_{t-1}^k[n_{t-1}] + F_\iota^a \mathbb{E}_{t-1}^k[a_t] \end{aligned}$$

Hence

$$\begin{aligned} \iota_t - \mathbb{E}_{t-1}^k[\iota_t] &= F_\iota^n(n_{t-1} - \mathbb{E}_{t-1}^k[n_{t-1}]) + F_\iota^a(\hat{a}_t^k - \mathbb{E}_{t-1}^k[a_t]) \\ &= F_\iota^n(h_{t-1} - \hat{h}_{t-1}^k) + F_\iota^a((I - G^k H)\varrho J^k(h_{t-1} - \hat{h}_{t-1}^k) + G^k(z_t^k - H\varrho\hat{a}_{t-1}^k)) \\ &= \left[F_\iota^n + F_\iota^a(I - G^k H)\varrho J^k\right](h_{t-1} - \hat{h}_{t-1}^k) + F_\iota^a G^k(z_t^k - H\varrho\hat{a}_{t-1}^k) \end{aligned} \tag{A42}$$

Since

$$\begin{aligned} F_\iota^a G^k(z_t^k - H\varrho\hat{a}_{t-1}^k) &= F_\iota^a G^k(Ha_t + w_t^k - H\varrho\hat{a}_{t-1}^k) \\ &= F_\iota^a G^k(H\varrho a_{t-1} + H\mu_t + w_t^k - H\varrho\hat{a}_{t-1}^k) \\ &= \underbrace{F_\iota^a G^k(H\mu_t + w_t^k)}_{\text{orthogonal to } h_{t-1}} + \underbrace{F_\iota^a G^k H\varrho(a_{t-1} - \hat{a}_{t-1}^k)}_{\text{correlates with } h_{t-1} - \hat{h}_{t-1}^k} \end{aligned}$$

we have

$$\begin{aligned} \iota_t - \mathbb{E}_{t-1}^k[\iota_t] &= \left[F_\iota^n + F_\iota^a(I - G^k H)\varrho J^k\right](h_{t-1} - \hat{h}_{t-1}^k) + F_\iota^a G^k(H\varrho a_{t-1} + H\mu_t + w_t^k - H\varrho\hat{a}_{t-1}^k) \\ &= \left[F_\iota^n + F_\iota^a(I - G^k H)\varrho J^k\right](h_{t-1} - \hat{h}_{t-1}^k) + F_\iota^a G^k H\varrho(a_{t-1} - \hat{a}_{t-1}^k) + F_\iota^a G^k(H\mu_t + w_t^k) \end{aligned} \tag{A43}$$

A.6 Proof of Proposition 3

The first part of the proposition follows from the context.

For the second part: without adjustment costs ($\xi_k = \xi_n = 0$), we have

$$k_{it} = n_{it} = (\epsilon - 1)\mathbb{F}_{it}[a_{it}]$$

hence $\iota_{it} = h_{it}$ and the regression coefficients $\tilde{\beta}_h = \tilde{\beta}_\iota$ and should always have the same sign.

Now I derive the sign result. The investment forecast error is

$$\begin{aligned}
\iota_{it} - \mathbb{F}_{i,t-1}[\iota_{it}] &= (\epsilon - 1)\left[\mathbb{F}_{it}[a_{it}] - \mathbb{F}_{i,t-1}[a_{it}]\right] \\
&= (\epsilon - 1)G(\rho a_{i,t-1} - \rho\lambda\mathbb{F}_{i,t-1}[a_{i,t-1}]) + (\epsilon - 1)G(\mu_{it} + \varepsilon_{it}) \\
&= (\epsilon - 1)G\rho(a_{i,t-1} - \mathbb{F}_{i,t-1}[a_{i,t-1}]) + (\epsilon - 1)G\rho(1-\lambda)\mathbb{F}_{i,t-1}[a_{i,t-1}] + (\epsilon - 1)G(\mu_{it} + \varepsilon_{it}) \\
&= \epsilon G\rho(y_{i,t-1} - \mathbb{F}_{i,t-1}[y_{i,t-1}]) + (\epsilon - 1)G\rho(1-\lambda)\mathbb{F}_{i,t-1}[a_{i,t-1}] + (\epsilon - 1)G(\mu_{it} + \varepsilon_{it})
\end{aligned} \tag{A44}$$

where the last step follows from

$$y_{i,t-1} - \mathbb{F}_{i,t-1}[y_{i,t-1}] = (1 - 1/\epsilon)(a_{i,t-1} - \mathbb{F}_{i,t-1}[a_{i,t-1}])$$

The investment $\iota_{i,t-1}$ is given by

$$\iota_{i,t-1} = k_{i,t-1} - k_{i,t-2} = (\epsilon - 1)(\mathbb{F}_{i,t-1}[a_{i,t-1}] - \mathbb{F}_{i,t-2}[a_{i,t-2}]) \tag{A45}$$

The signs of regression coefficients $\tilde{\beta}_h$ or $\tilde{\beta}_\iota$ are determined by

$$\begin{aligned}
sgn(\tilde{\beta}_h) = sgn\Big[&\mathbb{E}[(y_{i,t-1} - \mathbb{F}_{i,t-1}[y_{i,t-1}])^2]\mathbb{E}[\iota_{i,t-1}(\iota_{it} - \mathbb{F}_{i,t-1}[\iota_{it}])] \\
&- \mathbb{E}[(y_{i,t-1} - \mathbb{F}_{i,t-1}[y_{i,t-1}])\iota_{i,t-1}]\mathbb{E}[(y_{i,t-1} - \mathbb{F}_{i,t-1}[y_{i,t-1}])(\iota_{it} - \mathbb{F}_{i,t-1}[\iota_{it}])]\Big]
\end{aligned} \tag{A46}$$

To save some notation, define $a^{NE} \equiv a_{i,t-1} - \mathbb{F}_{i,t-1}[a_{i,t-1}]$, $\Delta F \equiv F_{i,t-1}[a_{i,t-1}] - \mathbb{F}_{i,t-2}[a_{i,t-2}]$, and $F \equiv \mathbb{F}_{i,t-1}[a_{i,t-1}]$. Hence we have

$$\mathbb{E}[(y_{i,t-1} - \mathbb{F}_{i,t-1}[y_{i,t-1}])^2] = (1 - 1/\epsilon)^2 var(a^{NE})$$

$$\mathbb{E}[\iota_{i,t-1}(\iota_{it} - \mathbb{F}_{i,t-1}[\iota_{it}])] = (\epsilon - 1)^2 G\rho \cdot cov(\Delta F, a^{NE}) + (\epsilon - 1)^2 \rho G(1-\lambda) \cdot cov(\Delta F, F)$$

$$\mathbb{E}[(y_{i,t-1} - \mathbb{F}_{i,t-1}[y_{i,t-1}])\iota_{i,t-1}] = (1 - 1/\epsilon)(\epsilon - 1) \cdot cov(a^{NE}, \Delta F)$$

$$\mathbb{E}[(y_{i,t-1} - \mathbb{F}_{i,t-1}[y_{i,t-1}])(\iota_{it} - \mathbb{F}_{i,t-1}[\iota_{it}])] = (1-1/\epsilon)(\epsilon-1)\rho G var(a^{NE}) + (1-1/\epsilon)(\epsilon-1)G\rho(1-\lambda) \cdot cov(a^{NE}, F)$$

Hence the right hand side of (A46) is equal to

$$sgn\Big\{(1 - 1/\epsilon)^2(\epsilon - 1)^2 G\rho(1-\lambda)\underbrace{\left[cov(\Delta F, F)var(a^{NE}) - cov(a^{NE}, F)cov(a^{NE}, \Delta F)\right]}_{\equiv A}\Big\}$$

Since $(1-1/\epsilon)^2(\epsilon-1)^2 G\rho > 0$, we have

$$sgn(\tilde{\beta}_h) = sgn((1-\lambda)A) \tag{A47}$$

Now we pin down the sign of A. We know that

$$var(a) = \frac{\sigma_\mu^2}{1-\rho^2}$$

$$cov(F,a) = \frac{G\sigma_\mu^2}{(1-\rho^2)(1-\rho^2\lambda(1-G))}$$

$$var(F) = \frac{G^2\left[(1+\lambda\rho^2(1-G))\sigma_\mu^2 + (1-\rho^2)(1-\rho^2\lambda(1-G))\sigma_\varepsilon^2\right]}{(1-\rho^2)(1-\rho^2\lambda(1-G))(1-\rho^2\lambda^2(1-G)^2)}$$

Hence

$$\begin{aligned} A &= (1-\rho\lambda(1-G))var(F)var(a) - \rho(1-\lambda(1-G))cov(a,F)var(F) \\ &\quad - \rho G\cdot cov(F,a)var(a) + (\rho G - 1 + \rho)cov(a,F)^2 \\ &= \frac{\sigma_\mu^2\left[P\sigma_\mu^2 + Q\sigma_\varepsilon^2\right]}{(1-\rho^2)^2(1-\rho^2\lambda(1-G))^2(1-\rho^2\lambda^2(1-G)^2)} \end{aligned} \tag{A48}$$

where

$$P = (1-\rho^2)\rho\lambda(\rho\lambda-1)(1-G)^2G^2$$

and

$$Q = (1-\rho^2)(1-\rho^2\lambda(1-G))((1-\rho\lambda(1-G))(1-\rho^2\lambda(1-G)) - \rho G(1-\lambda(1-G)))$$

The sign of A is the same as $P\sigma_\mu^2 + Q\sigma_\varepsilon^2$:

$$A > 0 \Leftrightarrow P\sigma_\mu^2 + Q\sigma_\varepsilon^2 > 0$$

$$\Longrightarrow \frac{\sigma_\mu^2}{\sigma_\varepsilon^2} < \frac{(1-\rho^2\lambda(1-G))\left[(1-\rho\lambda(1-G))(1-\rho^2\lambda(1-G)) - \rho G(1-\lambda(1-G))\right]}{\rho\lambda(1-\rho\lambda)(1-G)^2G^2} \tag{A49}$$

If (A49) is satisfied, then $sgn(\tilde{\beta}_h) = sgn(\tilde{\beta}_\iota) = sgn(1-\lambda)$.

A.7 Proof of Propositions in Section 4

I first show that the FOCs in the MPBE can be expressed in the form of (32) and (33), as in Kaymak and Schott (2023). Define

$$\begin{aligned}\omega_{it}^k &= -\frac{\xi_k}{\bar{R}}(k_{it} - k_{i,t-1}) + \beta\frac{1}{\bar{R}\bar{K}}\mathbb{E}_{it}^k\left[\frac{\partial V_{i,t+1}^k}{\partial k_{it}}\right] \\ &= -\frac{\xi_k}{\bar{R}}(k_{it} - k_{i,t-1}) + \beta\frac{1}{\bar{R}\bar{K}}\left[(\hat{P}_{12}^k + \hat{P}_{21}^k) + 2\hat{P}_{22}^k k_{it} + (\hat{P}_{32}^k + \hat{P}_{23}^k)\hat{n}_{it}^k + (\hat{P}_{42}^k + \hat{P}_{24}^k)\rho\hat{a}_{it}^k\right]\end{aligned} \tag{A50}$$

and

$$\begin{aligned}\omega_{it}^n &= -\frac{\xi_n}{\bar{W}}(n_{it} - n_{i,t-1}) + \beta\frac{1}{\bar{W}\bar{N}}\mathbb{E}_{it}^n\left[\frac{\partial V_{i,t+1}^n}{\partial n_{it}}\right] \\ &= -\frac{\xi_n}{\bar{W}}(n_{it} - n_{i,t-1}) + \beta\frac{1}{\bar{W}\bar{N}}\left[(\hat{P}_{13}^n + \hat{P}_{31}^n) + 2\hat{P}_{33}^n n_{it} + (\hat{P}_{32}^n + \hat{P}_{23}^n)\hat{k}_{it}^n + (\hat{P}_{43}^n + \hat{P}_{34}^n)\rho\hat{a}_{it}^n\right]\end{aligned} \tag{A51}$$

Plugging them into (32) and (33) gives us the FOCs consistent with an MPBE.

Next, we treat (32) and (33) as if it is a static problem: we characterize the decision rule for capital and labor as a function of ω^k and ω^n.

Hence k_{it} and n_{it} can be written as

$$k_{it} = -\tau_{it}^y + (\epsilon - 1)\hat{a}_{it}^k + (\hat{\alpha}_2 - 1)\epsilon\tau_{it}^k + (1 - \hat{\alpha}_2)\epsilon\omega_{it}^k + \hat{\alpha}_2\epsilon\omega_{it}^{(n,k)} \tag{A52}$$

$$n_{it} = -\tau_{it}^y + (\epsilon - 1)\hat{a}_{it}^n - \hat{\alpha}_1\epsilon\tau_{it}^k + (1 - \hat{\alpha}_1)\epsilon\omega_{it}^n + \hat{\alpha}_1\epsilon\omega_{it}^{(k,n)} \tag{A53}$$

where $\omega^{(n,k)}$ is the capital manager's perceived labor wedge and $\omega^{(k,n)}$ is the labor manager's perceived capital wedge:

$$\omega^{(k,n)} = -\frac{\xi_k}{\bar{R}}(\hat{k}_{it}^n - k_{i,t-1}) + \beta\frac{1}{\bar{R}\bar{K}}\left[(\hat{P}_{12}^k + \hat{P}_{21}^k) + 2\hat{P}_{22}^k\hat{k}_{it}^n + (\hat{P}_{32}^k + \hat{P}_{23}^k)n_{it} + (\hat{P}_{42}^k + \hat{P}_{24}^k)\rho\hat{a}_{it}^n\right] \tag{A54}$$

$$\omega^{(n,k)} = -\frac{\xi_n}{\bar{W}}(\hat{n}_{it}^k - n_{i,t-1}) + \beta\frac{1}{\bar{W}\bar{N}}\left[(\hat{P}_{13}^n + \hat{P}_{31}^n) + 2\hat{P}_{33}^n\hat{n}_{it}^k + (\hat{P}_{32}^n + \hat{P}_{23}^n)k_{it} + (\hat{P}_{43}^n + \hat{P}_{34}^n)\rho\hat{a}_{it}^k\right] \tag{A55}$$

For convenience, denote $\hat{\omega}^{(k,n)} \equiv \omega^{(k,n)} - \omega^k$ and $\hat{\omega}^{(n,k)} \equiv \omega^{(n,k)} - \omega^n$. We have

$$\hat{\omega}^{(k,n)} = \left[-\frac{\xi_k}{\bar{R}} F_\iota^a + \frac{\beta}{\bar{R}\bar{K}} \left(2\hat{P}_{22}^k F_\iota^a + (\hat{P}_{23}^k + \hat{P}_{32}^k) F_h^a + (\hat{P}_{24}^k + \hat{P}_{42}^k)\rho \right) \right] (\hat{a}^n - \hat{a}^k) \equiv A^{(k,n)} (\hat{a}^n - \hat{a}^k) \tag{A56}$$

and

$$\hat{\omega}^{(n,k)} = \left[-\frac{\xi_n}{\bar{W}} F_h^a + \frac{\beta}{\bar{W}\bar{N}} \left(2\hat{P}_{33}^n F_h^a + (\hat{P}_{23}^n + \hat{P}_{32}^n) F_\iota^a + (\hat{P}_{34}^n + \hat{P}_{43}^n)\rho \right) \right] (\hat{a}^k - \hat{a}^n) \equiv A^{(n,k)} (\hat{a}^k - \hat{a}^n) \tag{A57}$$

It is then trivial to prove the propositions.

Proof of Proposition 4 We take the difference between k and n decision rules above and the conclusion follows.

Proof of Proposition 5 By definition of MRPK and MRPN, we have

$$\begin{aligned} mrpk &= (1 - 1/\epsilon)a + (\hat{\alpha}_1 - 1)k + \hat{\alpha}_2 n \\ &= \tau^k + \tau^y + (\hat{\alpha}_1 - 1)(\epsilon - 1)(\hat{a}^k - a) + \hat{\alpha}_2(\epsilon - 1)(\hat{a}^n - a) - \omega^k + (\hat{\alpha}_1 - 1)\hat{\alpha}_2 \epsilon \hat{\omega}^{(n,k)} + \hat{\alpha}_1 \hat{\alpha}_2 \epsilon \hat{\omega}^{(k,n)} \\ &= \tau^k + \tau^y + A^k(\hat{a}^k - a) + A^n(\hat{a}^n - a) - \omega^k \end{aligned} \tag{A58}$$

where

$$A^k \equiv (\hat{\alpha}_1 - 1)(\epsilon - 1) + (\hat{\alpha}_1 - 1)\hat{\alpha}_2 \epsilon A^{(n,k)} - \hat{\alpha}_1 \hat{\alpha}_2 \epsilon A^{(k,n)}$$

$$A^n \equiv \hat{\alpha}_2(\epsilon - 1) - (\hat{\alpha}_1 - 1)\hat{\alpha}_2 \epsilon A^{(n,k)} + \hat{\alpha}_1 \hat{\alpha}_2 \epsilon A^{(k,n)}$$

and

$$\begin{aligned} mrpn &= \tau^y + \hat{\alpha}_1(\epsilon - 1)(\hat{a}^k - a) + (\hat{\alpha}_2 - 1)(\epsilon - 1)(\hat{a}^n - a) - \omega^n + (\hat{\alpha}_2 - 1)\hat{\alpha}_1 \epsilon \hat{\omega}^{(k,n)} + \hat{\alpha}_1 \hat{\alpha}_2 \epsilon \hat{\omega}^{(n,k)} \\ &= \tau^y + B^k(\hat{a}^k - a) + B^n(\hat{a}^n - a) - \omega^n \end{aligned} \tag{A59}$$

where

$$B^k \equiv \hat{\alpha}_1(\epsilon - 1) - (\hat{\alpha}_2 - 1)\hat{\alpha}_1 \epsilon A^{(k,n)} + \hat{\alpha}_1 \hat{\alpha}_2 \epsilon A^{(n,k)}$$

$$B^n \equiv (\hat{\alpha}_2 - 1)(\epsilon - 1) + (\hat{\alpha}_2 - 1)\hat{\alpha}_1 \epsilon A^{(k,n)} - \hat{\alpha}_1 \hat{\alpha}_2 \epsilon A^{(n,k)}$$

Taking variance on these expressions yields the conclusions in Proposition 5.

A.8 Robustness Checks

In this section, I consider several robustness checks to Section 3 and Section 4.

A.8.1 Time-to-build

In the capital adjustment literature, it is common to assume that the capital takes time to build (e.g. Cooper and Haltiwanger (2006), David and Venkateswaran (2019)): today's investment transforms into capital stock with a one-period lag. One may wonder whether time-to-build invalidates the power of empirical test: namely, whether incorporating time-to-build in capital alone makes the coefficient of interest $\tilde{\beta}_h \neq 0$.

I verify that time-to-build in capital does not invalidate the regression test. In our framework of Dynamic Narrow Framing, the only change is on the profit function: without time-to-build, we have

$$\Pi(K, N; A, K_{-1}, N_{-1}) = \underbrace{AK^{\alpha(1-1/\epsilon)}N^{(1-\alpha)(1-1/\epsilon)}}_{=\mathcal{R}(A,K,N)} - RK - WN - \text{adj. costs}$$

With time-to-build on capital, it seems that the only change is on the profit function:

$$\Pi(K, N; A, K_{-1}, N_{-1}) = AK_{-1}{}^{\alpha(1-1/\epsilon)}N^{(1-\alpha)(1-1/\epsilon)} - RK - WN - \text{adj. costs} \tag{A60}$$

In other words, with the time-to-build friction, capital for production today is predetermined, and the cost for tomorrow's capital is paid today. This friction changes the matrices in the log-quadratic profit function (A2): now the matrix $P, Q, R, H_\iota, H_h, H_{\iota h}$ becomes

$$P \equiv \begin{bmatrix} \frac{1}{2\epsilon}\bar{Y} & 0 & 0 & \frac{1}{2}(1-1/\epsilon)\bar{Y} \\ * & \frac{1}{2}\bar{R}\bar{K}(\alpha(1-1/\epsilon)-1) & \frac{1}{2}\alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} & \frac{1}{2}\alpha(1-1/\epsilon)^2\bar{Y} \\ * & * & \frac{1}{2}\bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) & \frac{1}{2}(1-\alpha)(1-1/\epsilon)^2\bar{Y} \\ * & * & * & \frac{1}{2}(1-1/\epsilon)^2\bar{Y} \end{bmatrix} \tag{A61}$$

$$Q \equiv \begin{bmatrix} -\bar{R}\bar{K} \\ -\bar{R}\bar{K} \\ 0 \\ 0 \end{bmatrix} \text{ and } R \equiv \begin{bmatrix} 0 \\ \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} \\ \bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) \\ (1-\alpha)(1-1/\epsilon)^2\bar{Y} \end{bmatrix} \tag{A62}$$

and

$$H_\iota \equiv -\frac{1}{2}(\bar{R}\bar{K} + \xi_k \bar{K}), H_h \equiv -\frac{1}{2}([1-(1-\alpha)(1-1/\epsilon)]\bar{W}\bar{N} + \xi_n \bar{N}) \text{ and } H_{\iota h} = 0 \quad \text{(A63)}$$

The learning process of firm managers is not changed by the time-to-build friction, and therefore the intuition we built in Section 3 remains true: lagged hiring cannot forecast the investment forecast error.

Table A1: Estimated Parameters: Baseline vs Time-to-build

Parameter		Model (Baseline)	Model (with time-to-build)
Convex Cost, Capital	ξ_k	1.84	0.17
Convex Cost, Labor	ξ_n	2.13	1.94
Noise on the common signal	$\sigma_{\epsilon,p}$	3.46	3.63
Noise on K manager's signal	$\sigma_{\epsilon,k}$	2.53	10.14
Noise on N manager's signal	$\sigma_{\epsilon,n}$	1.43	4.64
Dispersion of Capital Distortion	$\sigma_{\tau,k}$	1.09	0.13
Dispersion of Revenue Distortion	$\sigma_{\tau,y}$	0.37	0.36

Table A2: Targeted Moments: Baseline vs Time-to-build

Moment	Data	Model (Baseline)	Model (Non-convex Cost)
corr(ι, ι_{-1})	0.294	0.293	0.294
corr(h, h_{-1})	0.178	0.178	0.178
corr(ι, h)	0.526	0.526	0.526
$\tilde{\beta}_h$	0.086	0.086	0.086
$\tilde{\beta}_\iota$	-0.034	-0.034	-0.034
var(ι)	0.018	0.013	0.018
var(h)	0.020	0.023	0.020

Results

A.8.2 Reduced-form Distortions

I show that reduced-form distortions cannot reproduce the sign patterns by themselves. Consider reduced-form distortions as in Hsieh and Klenow (2009): the profit of the firm is given by

$$\Pi_{it} = (1-\tau^y_{it})R(K_{it}, N_{it}; A_{it}) - (1+\tau^k_{it})R_t K_{it} - W_t N_{it}$$

Table A3: Sources of Capital-labor Disconnect: Model with Time-to-build

	corr(ι, h)
Time to build	0.628
Convex Costs	0.993
- Capital Adjustment Cost ξ_k	0.939
- Labor Adjustment Cost ξ_n	0.892
Information frictions	0.564
- Public Signal $\sigma_{\epsilon,p}$	1.000
- Capital Manager's Private Signal $\sigma_{\epsilon,k}$	-0.028
- Labor Manager's Private Signal $\sigma_{\epsilon,n}$	-0.024
Transitory, Uncorrelated Distortion τ	0.996
- Transitory, Uncorrelated Capital Distortion $\sigma_{\tau,k}$	0.974
- Transitory, Uncorrelated Revenue Distortion $\sigma_{\tau,y}$	1.000
All mechanisms	0.526
Data	0.526

$$-\frac{\xi_k}{2}\left(\frac{K_{it}}{K_{i,t-1}}-1\right)^2 K_{i,t-1}-\frac{\xi_n}{2}\left(\frac{N_{it}}{N_{i,t-1}}-1\right)^2 N_{i,t-1} \tag{A64}$$

where τ_{it}^y is a revenue distortion and τ_{it}^k is a capital-specific distortion. I follow David and Venkateswaran (2019) and assume that these distortions have the following structure:

$$\tau_{it}^y = \gamma^y a_{it} + \delta_i^y + \eta_{it}^y \tag{A65}$$

$$\tau_{it}^k = \gamma^k a_{it} + \delta_i^k + \eta_{it}^k \tag{A66}$$

where $\gamma^y, \gamma^k > 0$ are the "correlated distortions" as in Bartelsman et al. (2013), δ^y, δ^k captures firm characteristics (such as management quality in Bloom et al. (2019)) and η_{it}^y and η_{it}^k are the uncorrelated part of distortion as in Hsieh and Klenow (2009).

Define $\tilde{\eta}_{it}^y \equiv \delta_i^y + \eta_{it}^y$ and $\tilde{\eta}_{it}^k \equiv \delta_i^k + \eta_{it}^k$. The state vector x_{it} is now defined as

$$x_{it} = [1, k_{i,t-1}, n_{i,t-1}, a_{it}, \tilde{\eta}_{it}^k, \tilde{\eta}_{it}^y]'$$

and the profit function can be approximated as

$$
\begin{aligned}
\Pi_{it} \approx\ & \frac{1}{\epsilon}\bar{Y} + \left(1-\frac{1}{\epsilon}\right)\bar{Y}a_{it} - \bar{Y}(\gamma^y a_{it} + \tilde{\eta}^y_{it}) - \bar{R}\bar{K}\left(\gamma^k a_{it} + \tilde{\eta}^k_{it}\right) - \bar{R}\bar{K}\left(\gamma^k a_{it} + \tilde{\eta}^k_{it}\right)k_{it} \\
& + \frac{1}{2}\left([\alpha(1-1/\epsilon)-1]\bar{R}\bar{K} - \xi_k\bar{K}\right)k_{it}^2 + \frac{1}{2}\left([(1-\alpha)(1-1/\epsilon)-1]\bar{W}\bar{N} - \xi_n\bar{N}\right)n_{it}^2 \\
& + \frac{1}{2}\left(1-\frac{1}{\epsilon}\right)^2\bar{Y}a_{it}^2 - \frac{\xi_k}{2}\bar{K}k_{i,t-1}^2 - \frac{\xi_n}{2}\bar{N}n_{i,t-1}^2 + \xi_k\bar{K}k_{it}k_{i,t-1} + \xi_n\bar{N}n_{it}n_{i,t-1} \\
& + \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y}k_{it}n_{it} + \alpha(1-1/\epsilon)^2\bar{Y}k_{it}a_{it} + (1-\alpha)(1-1/\epsilon)^2\bar{Y}n_{it}a_{it} \\
& - \left(1-\frac{1}{\epsilon}\right)\bar{Y}(\gamma^y a_{it} + \tilde{\eta}^y_{it})a_{it} - \bar{R}\bar{K}(\gamma^y a_{it} + \tilde{\eta}^y_{it})k_{it} - \bar{W}\bar{N}(\gamma^y a_{it} + \tilde{\eta}^y_{it})n_{it}
\end{aligned}
$$

Everything can be mapped into the primitives in Lemma 1:

$$
P = \begin{pmatrix}
\frac{1}{\epsilon}\bar{Y} & 0 & 0 & \frac{1}{2}\Big((1-\frac{1}{\epsilon})\bar{Y} - \bar{R}\bar{K}\gamma_k - \bar{Y}\gamma_y\Big) & -\frac{1}{2}\bar{R}\bar{K} & -\frac{1}{2}\bar{Y} \\
* & \frac{1}{2}\bar{R}\bar{K}\Big(\alpha(1-\frac{1}{\epsilon})-1\Big) & \frac{1}{2}\alpha(1-\alpha)(1-\frac{1}{\epsilon})^2\bar{Y} & \frac{\epsilon-1}{2\epsilon}\bar{R}\bar{K} - \frac{1}{2}\bar{R}\bar{K}(\gamma_k+\gamma_y) & -\frac{1}{2}\bar{R}\bar{K} & -\frac{1}{2}\bar{R}\bar{K} \\
* & * & \frac{1}{2}\bar{W}\bar{N}\Big((1-\alpha)(1-\frac{1}{\epsilon})-1\Big) & \frac{\epsilon-1}{2\epsilon}\bar{W}\bar{N} - \frac{1}{2}\bar{W}\bar{N}\gamma_y & 0 & -\frac{1}{2}\bar{W}\bar{N} \\
* & * & * & \frac{1}{2}\frac{(\epsilon-1)^2}{\epsilon^2}\bar{Y} - (1-\frac{1}{\epsilon})\bar{Y}\gamma_y & 0 & -\frac{1}{2}(1-\frac{1}{\epsilon})\bar{Y} \\
* & * & * & * & 0 & 0 \\
* & * & * & * & * & 0
\end{pmatrix}
$$

$$
Q = \begin{pmatrix}
0 \\
\bar{R}\bar{K}\big(\alpha(1-\frac{1}{\epsilon})-1\big) \\
\alpha(1-\alpha)(1-\frac{1}{\epsilon})^2\bar{Y} \\
\frac{\epsilon-1}{\epsilon}\alpha(1-\frac{1}{\epsilon})\bar{Y} - \bar{R}\bar{K}(\gamma_k+\gamma_y) \\
-\bar{R}\bar{K} \\
-\bar{R}\bar{K}
\end{pmatrix}, \quad
R = \begin{pmatrix}
0 \\
\alpha(1-\alpha)(1-\frac{1}{\epsilon})^2\bar{Y} \\
\bar{W}\bar{N}\big((1-\alpha)(1-\frac{1}{\epsilon})-1\big) \\
\frac{\epsilon-1}{\epsilon}(1-\alpha)(1-\frac{1}{\epsilon})\bar{Y} - \bar{W}\bar{N}\gamma_y \\
0 \\
-\bar{W}\bar{N}
\end{pmatrix}
$$

$$
H_\iota = \frac{1}{2}\bar{R}\bar{K}\Big(\alpha(1-\frac{1}{\epsilon})-1\Big) - \frac{1}{2}\xi_k\bar{K}, \quad H_{\iota h} = \alpha(1-\alpha)(1-\frac{1}{\epsilon})^2\bar{Y}, \quad H_h = \frac{1}{2}\bar{W}\bar{N}\Big((1-\alpha)(1-\frac{1}{\epsilon})-1\Big) - \frac{1}{2}\xi_n\bar{N}
$$

$$
A = \mathrm{diag}(1,1,1,\rho,0,0), \quad B = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}
$$

A.8.3 Private Benefit of Managers

The team production literature (Marschak and Radner (1972)) typically assumes that managers are cooperative and non-strategic: they choose their actions to jointly maximize a common objective, which is the firm's profits. Here I deviate from this assumption and allow capital and labor managers to have private benefits. Following the managerial empire building literature (e.g. Ebsim, Lian, Ma, Ottonello, and Perez (2025)), I model this managerial friction by assuming that the capital manager's payoff function is

$$\Pi_{it} + g(\xi k_{it})$$

where $g(x) = x^{\phi}$ captures the private benefit of the capital manager by diverting a fraction ξ of the capital stock to their career concerns. After doing log-quadratic approximation to the private benefit term g, the capital manager's payoff can be written as

$$x_t' P^k x_t + x_t' Q^k \iota_t + x_t' R h_t + H_\iota^k \iota_t^2 + H_{\iota h} \iota_t h_t + H_h h_t^2$$

where P^k, Q^k and H_ι^k are altered by the private benefit term. The policy function of the capital and labor manager's problem remains linear, and hence Lemma 1 and Lemma 3 continue to hold in this case. The decomposition of investment forecast error in Proposition 1 remains valid.

A.8.4 Non-convex Adjustment Cost

Under non-convex costs, the MPBE characterization in Lemma 3 is no longer valid because non-convex costs make the decision rules of investment and hiring non-Gaussian. As a result, in an MPBE, capital and labor managers can no longer treat the realizations of the other manager's past decisions as additional Gaussian signals in the Kalman filtering.

To avoid this technical barrier and make the results comparable to the baseline estimation, I compute a temporary equilibrium where the evolution of managers' beliefs follows the law of motion in Lemma 2. This temporary equilibrium should be regarded as an approximation rather than an exact characterization of the true MPBE in this dynamic team production game of incomplete information. This class of temporary equilibrium nests the MPBE characterization in Lemma 3 when the non-convex costs parameters are set to 0.

I compute the non-linear decision rules for capital and labor managers by standard value function iteration algorithm. The value functions $V^k(k_{t-1}, n_{t-1}, \hat{a}_t^k)$ and $V^n(k_{t-1}, n_{t-1}, \hat{a}_t^n)$ have kinks in presence of non-convex costs. To speed up the computation, I define "ex-ante" value function W^k and W^n as

$$W^k(k_{t-1}, n_{t-1}, \hat{a}_{t-1}^k) \equiv \mathbb{E}_{t-1}^k[V^k(k_{t-1}, n_{t-1}, \hat{a}_t^k)] \text{ and } W^n(k_{t-1}, n_{t-1}, \hat{a}_{t-1}^n) \equiv \mathbb{E}_{t-1}^n[V^n(k_{t-1}, n_{t-1}, \hat{a}_t^n)]$$

These ex-ante value functions smooth out the kinks in V^k and V^n in $\hat{a}$ dimension, and we can therefore use Chebyshev polynomials on $\hat{a}$ and linear splines on k and n dimension to approximate W^k and W^n. The backward iteration steps are carried out on W^k and W^n instead of V^k and V^n, which significantly speeds up the computation. The backward iteration steps can be sketched as follows:

1. Let the result of previous iteration be $W_{(s)}^k$ and $W_{(s)}^n$;
2. At each grid points $(k, n, \hat{a}^k)$ of W^k, compute the following branch values:
   - the value of inaction $\pi(k, n, \hat{a}^k) + \beta W_{(s)}^k(k, n, \hat{a}^k)$;
   - the value of adjusting capital only: $\max_{k^*} \pi(k^*, n, \hat{a}^k) + \beta W_{(s)}^k(k^*, n, \hat{a}^k)$;
   - the value of adjusting labor only: $\max_{n^*} \pi(k, n^*, \hat{a}^k) + \beta W_{(s)}^k(k, n^*, \hat{a}^k)$;
   - the value of adjusting both capital and labor: $\max_{k^*, n^*} \pi(k^*, n^*, \hat{a}^k) + \beta W_{(s)}^k(k^*, n^*, \hat{a}^k)$;
3. Take the max of the above four branch values, get the updated value function $W_{(s+1)}^k(k, n, \hat{a}^k)$ policy function $k_{(s+1)}^*(k, n, \hat{a}^k)$ and the capital manager's perceived labor choice $\hat{n}_{(s+1)}^k(k, n, \hat{a}^k)$;
4. Do step 2 and 3 to W^n, get the updated value function $W_{(s+1)}^n(k, n, \hat{a}^n)$ policy function $n_{(s+1)}^*(k, n, \hat{a}^n)$ and the labor manager's perceived capital choice $\hat{k}_{(s+1)}^n(k, n, \hat{a}^n)$;
5. Go back to step 1 if $|W_{(s+1)}^k - W_{(s)}^k|$ and $|W_{(s+1)}^n - W_{(s)}^n|$ is larger than tolerance.

After solving the policy functions, I simulate a panel of 1000 firms for 500 periods to compute the targeted moments in Table A5. The estimated parameters are tabulated in Table A4.

Results

Table A4: Estimated Parameters: Baseline vs Non-Convex Cost

Parameter		Model (Baseline)	Model (with Non-convex Cost)
Convex Cost, Capital	ξ_k	1.84	1.71
Convex Cost, Labor	ξ_n	2.13	2.18
Noise on the common signal	$\sigma_{\epsilon,p}$	3.46	3.57
Noise on K manager's signal	$\sigma_{\epsilon,k}$	2.53	1.81
Noise on N manager's signal	$\sigma_{\epsilon,n}$	1.43	0.97
Dispersion of Capital Distortion	$\sigma_{\tau,k}$	1.09	0.95
Dispersion of Revenue Distortion	$\sigma_{\tau,y}$	0.37	0.39
Non-convex Cost, Capital	χ_k	-	0.0003
Non-convex Cost, Labor	χ_k	-	0.0002

Table A5: Targeted Moments: Baseline vs Non-convex Cost

Moment	Data	Model (Baseline)	Model (Non-convex Cost)
$\text{corr}(\iota, \iota_{-1})$	0.294	0.293	0.293
$\text{corr}(h, h_{-1})$	0.178	0.178	0.178
$\text{corr}(\iota, h)$	0.526	0.526	0.526
$\tilde{\beta}_h$	0.086	0.086	0.086
$\tilde{\beta}_\iota$	-0.034	-0.034	-0.034
$\text{var}(\iota)$	0.018	0.013	0.012
$\text{var}(h)$	0.020	0.023	0.023
Inaction Rate: Investment	0.091	*0.075*	0.091
Inaction Rate: Hiring	0.091	*0.057*	0.091

Note: The numbers in Italic are untargeted.

B Data Appendix

This appendix shows the details of sample construction and variable definition for the expectation test and the calibration exercise.

B.1 Construction of IBES-Compustat Dataset

Prepare the CRSP-Compustat Dataset After downloading the CRSP-Compustat Annual Dataset from the WRDS website, I apply the following filters:

- Keep only the firms incorporated in the US (`fic = "USA"`);

Table A6: Sources of Capital-labor Disconnect: Model with Non-convex Costs

	corr(ι, h)
Convex Costs	0.949
- Capital Adjustment Cost ξ_k	0.684
- Labor Adjustment Cost ξ_n	0.872
Information frictions	-0.074
- Public Signal $\sigma_{\epsilon,p}$	1.000
- Capital Manager's Private Signal $\sigma_{\epsilon,k}$	-0.026
- Labor Manager's Private Signal $\sigma_{\epsilon,n}$	-0.029
Transitory, Uncorrelated Distortion τ	0.987
- Transitory, Uncorrelated Capital Distortion $\sigma_{\tau,k}$	0.929
- Transitory, Uncorrelated Revenue Distortion $\sigma_{\tau,y}$	1.000
Non-convex Costs χ	1.000
- Capital fixed cost χ_k	1.000
- Labor fixed cost χ_n	1.000
All mechanisms	0.526
Data	0.526

- Keep only the firms using USD as accounting units (`curncd = "USD"`);
- Drop the observations with missing NAICS codes;
- Drop the observations with 2-digit NAICS codes = 52 (finance) and = 22 (utilities);
- Drop the observations with negative sales;
- Drop the observations with negative employment;
- Drop the observations with missing capital expenditure;
- Drop the observations with missing sales.
- Drop the observations with missing employment.
- Remove the duplicate (LPEMNO, fyear) combinations.

I end up getting a CRSP-Compustat Annual sample with 81,937 firm-year observations from 2000 to 2024.

Prepare the IBES-Compustat Joint Sample After downloading the IBES Guidance Dataset from WRDS, I apply the following filters and imputations:

- Keep only the annual guidance (`pdicity = "ANN"`);
- Keep only the sales and capital expenditure guidances (`measure = "CPX" or "SAL"`);
- Keep only the observations with range guidance (`range_desc = 1`) or point estimates (`range_desc = 2`). For the firms that adopt range guidance, drop the observations with missing lower or upper bound, and impute the forecast/nowcast by the midpoint of the lower and upper bound (`(val_1 + val_2)/2`).
- Merge the IBES-CRSP bridge file to the IBES sample by matching the IBES ticker (firm identifier in IBES datasets). This step allows us to map the IBES firm identifier (IBES ticker) to CRSP-Compustat firm identifier (PERMNO).
- Drop all the guidances with forecasting horizon $>$ 12 months.
- Up to now, the observations in the IBES Guidance dataset are at firm-year-guidance level. Drop the observation if it is the only guidance posted by the firm in that year. We are left with the guidances posted by the firms that make multiple guidances in a fiscal year.
- Drop the guidances that (1) are not the first guidance posted by the firm in a fiscal year, or (2) are not the last guidance posted by the firm in a fiscal year.
- Now, We are left with only the first and the last guidance posted by a firm in a fiscal year. I use the first guidance as "forecast" and the last guidance as "nowcast" for each firm and fiscal year. Up to now, each observation in the IBES Guidance dataset is at firm-year level.
- Merge the IBES Guidance dataset to the CRSP-Compustat dataset above and remove the observations with (1) missing capital expenditure forecasts, (2) missing lagged sales nowcasts, or (3) missing lagged hirings.

Summary Statistics We end up with a IBES-Compustat merged sample. This merged sample spans over 2000 and 2024 and contains 9201 firm-year observations. This is a sample for gigantic firms in the US. Table B1 compares the average and median size of firms in the IBES-Compustat sample with the whole CRSP-Compustat sample between 2000 and

2024. We can see that the firms in our IBES-Compustat sample is about two times larger in sales and employment on average. Median sales in the IBES-Compustat sample is about five times larger than the CRSP-Compustat firms. Table B2 tabulates the share of total sales in the US GDP for the firms in the samples. We can see that the total sales of the firms in our IBES-Compustat sample take up about 20 percent of US GDP, depite the fact that it only keeps about 10 percent of the CRSP-Compustat sample.

		CRSP-Compustat (2000-2024)		
Variables	Units	Mean	Median	No. Obs
Sales	Millions USD	3512.01	366.11	81937
Capital Stock (PPEGT)	Millions USD	2464.81	140.32	78478
Capital Expenditure	Millions USD	210.11	11.59	81937
Employment	Thousands	10.78	1.18	81937
		IBES-Compustat (2000-2024)		
Variables	Units	Mean	Median	No. Obs
Sales	Millions USD	6544.89	1811.80	8455
Capital Stock (PPEGT)	Millions USD	3491.86	771.47	8450
Capital Expenditure	Millions USD	281.56	64.29	8455
Employment	Thousands	22.45	6.90	8455

Table B1: Summary Statistics (2000-2024)

	CRSP-Compustat Sample			
	2005	2010	2015	2020
Total Sales (Billions USD)	8812	10057	11957	13413
Sales/GDP	68%	67%	65%	63%
	IBES-Compustat Sample			
	2005	2010	2015	2020
Total Sales (Billions USD)	1666	2954	3534	4146
Sales/GDP	13%	20%	19%	19%

Table B2: Total Sales and Share of Sales in US GDP

Construction of Variables for the Expectation Test The key variables in the expectation tests in Section 3 are constructed as follows. For the forecast and nowcast errors, I use the the arc-percent difference between the forecast/nowcast value and the actual value, as in Bloom, Codreanu, and Fletcher (2025):

$$\text{CapEX Forecast Error}_{it} = \frac{\text{CapEX}_{it} - \mathbb{E}^k_{i,t-1}[\text{CapEX}_{it}]}{1/2(\text{CapEX}_{it} + \mathbb{E}^k_{i,t-1}[\text{CapEX}_{it}])}$$

and

$$\text{Sales Nowcast Error}_{it} = \frac{\text{Sales}_{it} - \mathbb{E}^k_{i,t}[\text{Sales}_{it}]}{1/2(\text{Sales}_{it} + \mathbb{E}^k_{i,t}[\text{Sales}_{it}])}$$

The arc-difference definition of forecast/nowcast errors has two advantages. First, it is bounded between $[-2, 2]$, so that outliers in the realized values and expectations would not affect the forecast/nowcast error and the regression results by much. Additionally, it avoids the problems of taking logs when the actual values or expectations are close to 0. This may not be a big issue for the large-firm samples like IBES or Compustat, and I verify that the regression results are robust to alternative definitions of forecast/nowcast errors (e.g. the log difference between predicted values and actual values).

The definition of hiring is standard. For firm i in year t, hiring h_{it} is defined as

$$h_{it} \equiv \log(\text{emp}_{it}) - \log(\text{emp}_{i,t-1})$$

where `emp` is the employment of the firm in CRSP-Compustat. Alternatively, an arc-difference measure of hiring is widely adopted in the literature as well (e.g. Davis and Haltiwanger (1992)). The regression results do not change with alternative definitions of hiring.

B.2 Construction of Compustat Sample for Calibration

After obtaining COMPUSTAT Fundamentals Annual dataset from WRDS, I applied the following filter:

1. Exclude the observations before 1998;
2. Exclude the firms incorporated or legally registered outside the US (filter by COMPUSTAT terminology `fic`);
3. Exclude the firms that do not use US dollar as their native currency (filter by COMPUSTAT terminology `curncd`);
4. Following the literature (e.g. De Loecker, Eeckhout, and Unger (2020), Flynn and Sastry (2020), Afrouzi, Drenik, and Kim (2023), among many others), exclude firms in the following sectors
   (a) financial sector (SIC code 6000-6999, or 2-digit NAICS 52);

(b) utility sector (SIC code 4900-4999, or 2-digit NAICS 22);

(c) non-operating sector (SIC code 9995).

It is well documented that financial firms are different from non-financial firms in many aspects (e.g. production function, business cycle properties, etc). The input and output prices of the utility sector is heavily regulated.

5. Exclude observations with missing SIC or NAICS codes;

6. Exclude observations with missing total assets (COMPUSTAT terminology `at`);

7. Exclude observations with acquisitions (COMPUSTAT terminology `aq`) exceeding 5 percent of total assets (COMPUSTAT terminology `at`);
   This is a common practice in the literature because firms that went through large mergers and acquisitions tend to have very large changes in input choices and measured productivity. [12]

8. Exclude observations with non-positive sales (COMPUSTAT terminology `sale`);

9. Exclude observations with non-positive employments (COMPUSTAT terminology `emp`);

10. Exclude observations with missing cost of goods sold.

12. Take the following steps to back out the capital share α:

    (a) merge compustat with County Business Pattern (CBP) from Census (which contains industry-level wages - here, industry = 3-digit NAICS for manufacturing and 2-digit for other industries);

    (b) fill up the missing values in wage bill for Compustat firms: wage bill = emp * CBP_{wages};

    (c) compute the material expenditure = cogs + xsga - dp - wage bill (see Keller and Yeaple (2009) and Flynn and Sastry (2020)).

    (d) compute value-added = sales - material expenditure;

    (e) compute the labor share in value-added:

$$\text{rshare}_{\text{labor}} = \sum(wagebill)/\sum(\text{value-added})$$

[12] I may slack this filter if one believes that major M&As are (exogenous) variations to the management style and hence information friction.

Here, I sum up all the firm-year observations to get an aggregate share; in an alternative exercise I do the summation up to the industry level and get an industry-specific labor share for all the industries;

(f) back out the capital share of income:

$$\alpha = 1 - rshare_{labor}/(1 - 1/\epsilon)$$

13. I prepare the all the variables that will be used in the quantitative analysis:

    (a) Define capital stock as the book value of PPE (PPEGT);

    (b) take log of value-added, employment and capital stock, get the lagged and first-differenced log employment and log capital stock;

    (c) Define the profit shock/Sales Solow residual as

    $$a = (1-1/\epsilon)^{-1}\left[\log(valueadded) - \alpha(1-1/\epsilon)\log(capitalstock) - (1-\alpha)(1-1/\epsilon)\log(emp)\right]$$

    (d) get the lagged and first-differenced series for a;

    (e) define mrpk = log(value added) - log(capital stock) and mrpn = log(value added) - log (emp).

14. I append the cleaned IBES dataset to the cleaned Compustat sample. Each firm-year observation in the cleaned IBES dataset has the current year's first guidance of capital expenditure and the previous year's last guidance of sales. Call the resulting sample "IBES/Compustat Merged Quantitative" Sample.

15. In this "IBES/Compustat Merged Quantitative" Sample, I regress $k, n, a, \Delta k, \Delta n, \Delta a, mrpk, mrpn$ on sector-year fixed effects and keep the residuals (this is to remove the trends in the data).

16. I trim the 1 percent tail of the above residuals;

17. I trim the 1 percent tails of investment forecast errors and sales nowcast errors.

This results in a sammple that has 6492 firm-year observations. Table B3 reports the summary statistics.

Table B3: Summary Statistics of the Quantitative Sample

Variables	Units	Mean	Median	No. Obs
Sales	Millions USD	5319.80	1769.03	6492
Capital Stock (PPEGT)	Millions USD	2260.60	734.10	6492
Capital Expenditure	Millions USD	186.60	61.00	6492
Employment	Thousands	16.74	6.45	6492